# Theorem 2 for lattice Yang–Mills theory: proof chapter, version 4

(shortest route; Route K reduction verified)

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# Preamble

Written 2026-10-28 (Opus). v4 = v3 (`THEOREM2_PROOF_v3.md`[1]) updated with the verified material that appeared after v3: `B2_TWOSTEP.md` + `B2_TWOSTEP_VERIFY.md`, `H2_LARGE_L.md` + `H2_LARGE_L_VERIFY.md`, `lean/README.md` (Lemma D″ in Lean), `BETA_FP_U.md` (Route B inputs), `numerics/NUMERICS.md` §7. Main changes: §5 is rewritten around (H2); §8, §9, §10 are updated; small consistency edits elsewhere. It does not edit v1, v2, or v3. It is not itself verified as a whole document. It adds no claim beyond the cited verifier verdicts. One deliberate tag change: the logic of Theorem B (Route B) is tagged [proved+verified, conditional] per `BETA_FP_PW_VERIFY` §6 items 1–4; see the change log. The change log is at the end (Appendix).

**Tag legend.**

- [proved+verified]: proved in a project file and confirmed by an independent verifier file.
- [proved+verified, conditional on $X$]: the implication is confirmed; $X$ is open or assumed.
- [proved, unverified]: proved in a project file; no verifier has graded it yet.
- [Lean]: formalized in Lean 4 + Mathlib, no `sorry`, axioms propext, Classical.choice, Quot.sound only.
- [verified quote]: read on the page image of Balaban's paper (`reports/BALABAN_EXCERPTS_CHECK.md` or a verifier file).
- [numerical, reproduced]: floating-point result reproduced by a verifier. [numerical, run by the verifier]: run once, by a verifier. [numerical]: not reproduced. None is certified.
- [argued]: a plausibility argument, not a proof. [open]: no proof. [refuted]: shown false.

[1] Project files named in this chapter live in the `notes/` directory of the `yang-mills` repository.

- [Balaban]: an input taken from Balaban's published papers, used as stated there.
- [not written]: a proof plan; no project file contains the proof.

**Range of $L$.** Balaban assumes $L$ odd, $L > 11$, as a standing assumption for the whole of CMP 109 (pp. 251–252) [8]. His construction of $\beta_j(g)$ for $g > 0$ and his inductive assumption (H0) are therefore available only at odd $L > 11$. Every statement below that uses (H0) or $\beta_j(g)$ at $g > 0$ is stated for odd $L > 11$, or carries "(H0) and Balaban's construction at this $L$" as an explicit hypothesis. Statements about the one-loop quantities $\beta_j(0)$, $S_k$, and the symbols $\sigma_j$ hold for odd $L \geq 3$.

# 1 Statement

## 1.1 Balaban's statements [verified quote]

Theorem 2 (CMP 109 (1987), p. 259): "Let $d = 4$, $G = SU(2)$, and let $\gamma$ be a sufficiently small positive constant, then for a sufficiently small positive $g$ there exists a bare coupling constant $g_0 = g_0(\varepsilon, g)$ such that the sequence of the effective coupling constants $g_k$ is contained in the interval $]0, \gamma]$, and $g_K = g$. Moreover, there exist constants $\beta, \beta'$, $0 < \beta \leq \beta'$, such that $g_k$ satisfy the inequality $1/g^2 + \beta \log(L^k \varepsilon)^{-1} \leq 1/g_k^2 \leq 1/g^2 + \beta' \log(L^k \varepsilon)^{-1}$. (0.31)" Balaban adds that the proof "will be given in a separate paper". It was never published (CMP 122 II p. 355, verbatim: "has not been published yet, so we have Theorem 1 with the assumption").

Theorem 1 (CMP 122 II p. 355): "If the sequence of the effective coupling constants is contained in an interval $]0, \gamma]$ with a sufficiently small positive $\gamma$, then the effective densities $\rho_k$ have the form, and satisfy all the conditions and bounds, described in Sect. 2 [III]." The bound (0.1) there reads $\chi_k \exp[-(1/g_k^2)A(U_k) - E_-|T_\eta|] \leq \rho_k \leq \exp E_+|T_\eta|$ (corrected reading, `BALABAN_EXCERPTS_CHECK` item 3).

*Setting.* $\varepsilon = L^{-K}$, so $\log(L^k\varepsilon)^{-1} = (K-k)\log L$. $u_k := 1/g_k^2$. Recursion (CMP 109 (0.18)/(0.20)): $u_{k+1} = u_k - \beta_{k+1}(g_k)$. The increment $\beta_{k+1}$ depends on the whole history $g_0, \ldots, g_k$ (CMP 109 p. 298, [verified quote]). $\beta_j(0)$ is the one-loop (Gaussian) increment; it is history-independent by definition. Balaban assumes $L$ odd, $L > 11$, for the whole paper (CMP 109 pp. 251–252, OCR line 124, verbatim confirmed in `THEOREM2_PROOF_VERIFY` Step 6).

## 1.2 What Balaban's Theorem 1 actually needs (`LEMMA_DPRIME_SWEEP` §2, §7)

The quoted hypothesis of Theorem 1 is only $g_k \in ]0, \gamma]$. The proof uses more. CMP 119 Theorem 1 (p. 262) assumes "(I.0.33)", a misnumbered reference to (0.31) of CMP 109. Its working form is CMP 119 p. 255–256 (2.6)–(2.9), for scales $m < n$:

- (2.6) $g_n \leq (1 + g_n^2 \beta'(n-m))^{1/2} g_m$ and $g_m \leq (1 + \beta_0) g_n$;
- (2.7) $(\log g_n^{-2})^p \leq (1+\beta_0)(\log g_m^{-2})^p$ and $(\log g_m^{-2})^p \leq (1 + g_n^2\beta'(n-m))^{\beta_0}(\log g_n^{-2})^p$;
- (2.8), (2.9) the same for $\varepsilon_j = g_j p_0(g_j)$ and $R_j$.

Here $\beta_0$ is Balaban's small constant (1/7 in CMP 122 II p. 389), not a coupling. [verified quote] for (2.6)–(2.9): page images, `LEMMA_DPRIME_SWEEP` §1; the sweep itself is single-author, see §2.3.

In $u$-variables these follow from two pairwise bounds:

- (P-up) $u_m \leq u_n + B'(n-m)$, with no additive constant;

- (P-low) $u_m \geq u_n + \beta(n-m) - D'$, with $\beta > 0$, $D' \geq 0$ fixed,

plus the smallness condition

- (S) $\gamma^{-2} \geq c_{sw} D'$, $c_{sw} = \max(2, 1/(1-(1+\beta_0)^{-2})) = 64/15$ at $\beta_0 = 1/7$, and $\gamma \leq \gamma_*(\beta_0, \beta, B', p_0, p_1, r, \nu, \kappa_0)$ (the conditions that literal (0.31) also needs). $\gamma_*$ is free of the $D'$-ratio conditions, but it depends on $R$ through $B'$ (it contains $\gamma^2 B' \leq (1+\beta_0)^2 - 1$ with $B' = 5b/4 + 2R$). The order of constants in §6 ($R$ before $\gamma$) handles this.

$\beta_0$ must be fixed before $\gamma$.

### 1.3 The weak form proved conditionally in this chapter

**Theorem** (Theorem 2w (weak form)). *Let $G = SU(2)$, $d = 4$, $L$ odd, $L > 11$ (equivalently: $L$ odd $\geq 3$ with (H0) and Balaban's construction of $\beta_j(g)$ at this $L$ as an explicit hypothesis). There are $\gamma > 0$, $g_1 \in ]0, \gamma]$, $\beta > 0$, $B' \geq \beta$, $D' \geq 0$ such that for every $K \geq 1$ and every $g \in ]0, g_1]$ there is a bare coupling $g_0 = g_0(K, g)$ with*

- *(b) $g_k \in ]0, \gamma]$ for $0 \leq k \leq K$, and $g_K = g$;*
- *(c′) for all $0 \leq m \leq n \leq K$: $u_n + \beta(n-m) - D' \leq u_m \leq u_n + B'(n-m)$;*
- *(S) holds for $\gamma$, $D'$.*

*Consequence: Balaban's (0.31) holds up to the additive constant $D'$ on the lower side, with constants $\beta/\log L$ and $B'/\log L$. The two-sided bound fixes $1/g_k^2$ up to constants, as required.*

Status: [proved+verified, conditional on (B2) and the fixed-level inputs of §4] by Route K (Theorem K + Lemma D″). (B2) is reduced to the single inequality (H2) of §5 [proved+verified reduction: `B2_TWOSTEP_VERIFY` grade 3]. Second route: Route B (`BETA_FP_PW` Theorem 3.2 + Lemma D″). The logic of Theorem B is [proved+verified, conditional] (`BETA_FP_PW_VERIFY` §6 items 1–4). Its inputs (U), (DF), (T1) from `BETA_FP_U` are [proved, unverified]; the route is conditional on (U), ($B\theta$), and the plane-wave defects. `BETA_FP_U` reduces (U) to (T1) + (H2) [proved, unverified]. Both routes also need (H0) at this $L$ and L5 (§2.4). See §4.

### 1.4 Literal (0.31)

- (L1) Literal (0.31) for every $K \geq K_0$ and every $g \in ]0, \gamma]$. Needs eventual uniform positivity: there are $J$ and $\beta_* > 0$ with $\beta_j(0) \geq \beta_*$ for all $j > J$, plus $\sup_j |\beta_j(0)| < \infty$ (Lemma D′ (d), §2.1). [proved+verified; Lean] as an implication. Theorem K and Theorem B do NOT give eventual positivity: they bound window sums, not single increments. Eventual positivity follows from the convergence route (§7), which is open.

- (L2) Literal (0.31) for every $K \geq 1$. Needs $\inf_j \beta_j(0) > 0$ and $\sup_j \beta_j(0) < \infty$ (Lemmas D + E). If $\beta_j(0) \to \beta_\infty > 0$, then "every $\beta_j(0) > 0$" is equivalent to $\inf_j \beta_j(0) > 0$ (Lemma D′ Step 8). [proved+verified; Lean] as an implication. Inputs open.

- Literal (0.31) is not needed for Theorem 1: §1.2 shows the proof uses (c′) + (S). Literal (0.31) is used directly in one place only, the alternative choice of $\varepsilon_0$ in CMP 119 p. 248, which Balaban does not adopt.

- Necessity. Literal (0.31) at every $K \geq 1$ forces $\beta_K(0) \geq \beta \log L > 0$ for every $K$ (window $k = K-1$, let $g \to 0$ using (H0); so $L$ odd $> 11$). [proved, unverified]; elementary; `THEOREM2_PROOF_VERIFY` Point 5(a) agrees. So certified early steps are needed only for (L2), not for Theorem 1.

## 2 Shooting lemmas

Notation: $u_k = 1/g_k^2$, $t = 1/g^2$, $T(k,K) := \sum_{i=k+1}^{K} \beta_i(g_{i-1})$. Clamp $\varphi(u) := \max(u, 1/\gamma^2)^{-1/2}$. Lemmas D′, D″, D, E are abstract statements about a sequence of functions $\beta_j$. Applied to Balaban's $\beta_j(g)$, they need $L$ odd $> 11$ (or (H0) at this $L$) and L5.

### 2.1 Lemma D′ (shooting without early positivity) [proved+verified; Lean]

Sources: `LEMMA_DPRIME.md` §1–§2; `LEMMA_DPRIME_VERIFY.md` grades 1, 2, 5 (CONFIRMED). Step 7 of that file also holds grades 3, 4, 6, which are CORRECTED (Remark 1 overclaim, misreads in §5, ledger); this chapter uses none of them except through the sweep (§2.3). Lean: `lean/YmLean/LemmaDPrime.lean` (`YmLean.lemma_DPrime`, 405 lines), statement check `lean/VERIFY_LemmaDPrime.md`.

**Lemma** (Lemma D′)**.** Hypotheses. *(i)* $\beta_{k+1}$ *continuous on* $[0,\gamma_0]$ *and* $|\beta_{k+1}(g) - \beta_{k+1}(0)| \le C_1 g$, $C_1$ *independent of* $k$, $K$. *(ii)* $\sup_j |\beta_j(0)| \le B$. *(iii′) there is* $J$ *with* $\beta_j(0) \ge \beta_\infty/2$ *for all* $j > J$ *(this is all the proof uses of "*$\beta_j(0) \to \beta_\infty > 0$*").*

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, \beta_\infty/(4C_1))$; $B' := B + \beta_\infty/4$; $D := J \cdot B'$; $D' := J(B + \beta_\infty/2)$; $g_1 := (1/\gamma^2 + D)^{-1/2}$; $K_0 := J + \lceil 8D'/\beta_\infty \rceil$.

Conclusions.

- *(a)* $T(k,K) \ge -D$ *for all* $k < K$ *along any couplings in* $]0,\gamma]$.
- *(b) For every* $K \ge 1$ *and* $g \in ]0,g_1]$ *there is* $g_0$ *with all* $g_k \in ]0,\gamma]$ *and* $g_K = g$.
- *(c)* $u_K + (\beta_\infty/4)(K-k) - D' \le u_k \le u_K + B'(K-k)$. *The upper side has no constant.*
- *(d) For* $K \ge K_0$*: literal (0.31) with lower constant* $\beta_\infty/8$*, and shooting for every* $g \in ]0,\gamma]$. *If every* $\beta_j(0) > 0$*: literal (0.31) for all* $K$ *(Step 8 = Lemmas D + E).*

*Proof idea.* Clamped flow $\tilde{u}_{k+1} = \tilde{u}_k - \beta_{k+1}(\varphi(\tilde{u}_k))$ is continuous in $u_0$. Its tail sums lie in $[-D, B'(K-k)]$ for every $u_0$, because clamped couplings lie in $]0,\gamma]$. IVT on $[t-D, t+B'K]$ gives $\tilde{u}_K = t$. Then $\tilde{u}_k \ge t - D \ge 1/\gamma^2$ for $g \le g_1$, so the clamp is never active. The bound is derived for the clamped flow first; there is no circularity. For history-dependent $\beta$ this proof needs $\beta_{k+1}$ defined and continuous on the whole cube $]0,\gamma]^{k+1}$; see L5 in §2.4.

*Lean.* The Lean hypotheses are weaker (ContinuousOn instead of $C^1$; (iii′) instead of convergence). Step 8 is `lemma_DPrime_step8` (`lean/YmLean/LemmaDE.lean`). Lean covers one-argument $\beta_j(g)$ only, not the history form.

*Remark* (Remark 4). Without $g \le g_1$, (b) can fail at small $K$ ($\beta_1 = \beta_2 = \beta_3 = -0.8$, $K = 3$, $g = \gamma$). [proved+verified]

### 2.2 Lemma D″ (tail-sum form; no index $J$) [proved+verified; Lean]

Source: `KAPPA_BOUNDED.md` §5.3; `KAPPA_BOUNDED_VERIFY` §6 (CONFIRMED: "No $J$ needed"); hand rederivation in `THEOREM2_PROOF_v2_VERIFY` §A.

**Lemma** (Lemma D″)**.** Hypotheses. *(i) as in 2.1, and the window bound*

$$\textit{(5.1)} \quad T_0(k,K) := \sum_{i=k+1}^{K} \beta_i(0) \in [\, b(K-k) - 2R,\ b(K-k) + 2R \,] \quad \textit{for all } 0 \le k < K,$$

*with $b := c \log L > 0$ and some $R \geq 0$. (Route K gives $R = \kappa_*$; Route B gives $R = C_B$.)*

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, b/(4C_1))$; $\beta := 3b/4$; $B' := 5b/4 + 2R$; $D = D' := 2R$; $g_1 := (1/\gamma^2 + D)^{-1/2}$.

Conclusions. *(b) of 2.1, and the pairwise bound*

$$(c') \quad u_n + \beta(n-m) - D' \leq u_m \leq u_n + B'(n-m), \quad 0 \leq m \leq n \leq K,$$

*upper side without constant.*

*Proof.* $|T - T_0| \leq C_1\gamma(K-k) \leq (b/4)(K-k)$. Add (5.1). Steps 2–5 of Lemma D′ use only continuity and the bracket $\tilde{T} \in [-D, B'(K-k)]$; this bracket now comes from (5.1). The pairwise form is the same window argument with $K \to n$. The proof is the clamped-flow proof of 2.1; for history-dependent $\beta$ the L5 caveat of §2.4 applies.

Lean: `lean/YmLean/LemmaDDoublePrime.lean` (`YmLean.lemma_DDoublePrime`, 146 lines, no `sorry`, axioms propext, Classical.choice, Quot.sound; `lean/README.md`). It proves (a″), (b) with the clamp inactive on the solution, (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, and (d) below. The Lean hypotheses are weaker than the note: continuity instead of $C^1$; any $\gamma$ with $0 < \gamma \leq \gamma_0$ and $C_1\gamma \leq b/4$; $R \geq 0$ derived; $K = 0$ allowed; (c′) and (d) hold for every trajectory with couplings in $]0, \gamma]$. One-argument $\beta$ only (L5 not formalized).

(d) [Lean]: for $n - m \geq 8R/b$, $u_n + (b/2)(n-m) \leq u_m \leq u_n + B'(n-m)$. So literal (0.31) from (5.1) alone holds, with lower constant $b/2$, only for windows with $n - m \geq 8R/b$; short windows near $K$ are not controlled (see §1.4).

### 2.3 Lemma D″ meets Balaban's proof: the cross-scale sweep [proved, unverified (sweep); 4 of 19 table rows verified]

Source: `LEMMA_DPRIME_SWEEP.md` (all 12 Balaban files; OCR grep plus page image for every non-single-scale hit). `LEMMA_DPRIME_VERIFY` Steps 4–5 checked comparisons A–E on page images (CMP 122 I p. 192, p. 198 (1.94); II p. 381 ×2, p. 386) and corrected two misreadings of A (factor $(1+\beta_0)^2(k-j)^{\beta_0}$; $\delta'_j = g_j A_1 p_1(g_j)$). The five comparisons map to four table rows: A = row 11, E = row 12, C and D = row 16, B = row 18.

*Result.* 19 cross-scale uses of the coupling sequence (table in SWEEP §7), all in CMP 119 and CMP 122 I–II; none in CMP 95, 96, 98, 99, 102, 109, 116. Each is covered by (b) + (c′) + (S) of §1.2:

- items 1, 7, 10, 12, 17 need (P-up) with no constant (e.g. (2.6) line 1 is exactly (P-up));
- items 2, 3, 11, 13, 15, 16, 18 need (P-low) with fixed multiplicative slack $(1+\beta_0)$ or $(1+\beta_0)^2$; item 2 ((2.6) line 2) gives the constant 64/15; item 11 needs only $2.42D'$;
- items 4 ((2.8)–(2.9), $\varepsilon$ and $R$ ratios), 6 (CMP 119 (3.7), (3.8), (3.12), (3.13)), and 9 (CMP 122 I (1.52)), the last two of the form $\varepsilon_{k+1} \leq (1+\beta_0)\varepsilon_k$, use both sides; they follow as items 1 and 3 do (item 4 from items 1–3);
- item 14 (CMP 122 I p. 200, $R_{k-N_0+1}$ vs $R_k$) uses the lower side; it follows as item 3 does;
- item 5 (CMP 119 (2.31), (2.46): $\sum_j g_j^{\kappa_0} \leq g_n^{\kappa_0 - 6}$) needs linear growth (P-low) and $\gamma^{-2} \geq 2D'$;
- item 0 is CMP 119 Theorem 1 assuming (0.31); its proof works with (2.6)–(2.9);
- item 8 (CMP 119 p. 284) is in a sketch, not load-bearing.

Caveats [open]: (1) the sweep is a grep plus page images, not a line-by-line read; (2) "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) was not located; (3) Balaban says $\beta_0$ "can be chosen arbitrarily small, if $g$ is sufficiently small"; under (P-low) the floor is $\beta_0 \gtrsim D\gamma^2/2$, so $\beta_0$ must be fixed first; no step was found that needs $\beta_0 \to 0$; (4) the sweep has no independent verifier beyond rows 11, 12, 16, 18; (5) `BALABAN_EXCERPTS_CHECK` item 11 lists readings not checked on page images: II (4.37)/(5.1), CMP 119 (2.42), the (2.12)/(2.23) ambiguity, the "$L$ odd" page, the Sect. 3 estimates, and the 1989 bound lists.

### 2.4 Lemma D and Lemma E [proved+verified; Lean]

Sources: `LEMMAS.md` (corrected), `LEMMAS_DE_VERIFY.md` §1.4, §1.5, §2.4; `lean/YmLean/LemmaDE.lean` (`lemma_D`, `lemma_E`, `lemma_DE`).

- **Lemma D.** If $b \le \beta_j(g) \le b'$ on $]0,\gamma]$ with $b > 0$, then for every $K$ and $g \in ]0,\gamma]$ shooting succeeds and $t + b(K-k) \le u_k \le t + b'(K-k)$. This is literal (0.31). The first IVT interval in the draft was wrong and was refuted by the verifier; the corrected interval is $[t + bK,\ t + b'K]$. The nested-interval variant for $\beta$ defined only on admissible histories (`LEMMAS_DE_VERIFY` §1.5) is [proved+verified], not in Lean.

- **Lemma E.** If $\beta_0 \le \beta_j(0) \le \beta_0'$ and $|\beta_j(g) - \beta_j(0)| \le Cg^2$, then for $\gamma \le \sqrt{\beta_0/(2C)}$: $\beta_0/2 \le \beta_j(g) \le \beta_0' + \beta_0/2$. The verifier checked this $Cg^2$ form; the $C^1$ form ($C_1 g$, $\gamma \le \beta_0/(2C_1)$) is the same two-line proof [proved, trivial variant]. Lean has the $Cg^2$ form.

- **History dependence (ledger item L5)** [open]. Balaban's $\beta_{k+1}$ depends on $(g_0, \dots, g_k)$. Lemma D′ Remark 3 states the cube-continuity version (same proof). The gap is sharper than v2 stated (`THEOREM2_PROOF_v2_VERIFY` §F):

  - *Domain.* Balaban defines $\beta_{k+1}$ only on histories for which his Theorem 1 induction has run to step $k$ (`LEMMAS_DE_VERIFY` §2 item 1). By the sweep (§2.3), that induction uses (2.6)–(2.9), that is (c′) + (S), on the history, not only $g_i \in ]0,\gamma]$.
  - *The new gap.* The clamped shooting of §2.1–§2.2 evaluates $\beta$ on clamped histories $\tilde{g}_i = \varphi(\tilde{u}_i)$. A clamped history can violate the growth side of (c′): from $\max(a,c) - \max(b,c) \ge \min(a-b, 0)$ one gets only $u_m \ge u_n - D'$, not $u_m \ge u_n + \beta(n-m) - D'$. So the clamped shooting evaluates $\beta$ on histories outside Balaban's domain. The clamp proof is valid only under the cube hypothesis, which Balaban does not supply.
  - *A variant that avoids it* [not written]. Use a nested-interval proof in the style of `LEMMAS_DE_VERIFY` §1.5, with domain $D_k = \{u_0, \dots, u_k \ge \gamma^{-2}\}$ of true (unclamped) trajectories. On such trajectories, (c′) up to step $k$ follows from the tail-sum bound (a″) of Lemma D″, and (S) holds by the choice of $\gamma$. So every evaluation of $\beta$ stays inside Balaban's domain. At each step $k$, keep the set of $u_0$ whose trajectory stays in $D_k$; show by IVT, step by step, that this set contains a closed interval on which $u_k$ sweeps a range that contains the next target interval. For Lemma D (all increments positive) this is written. For Lemma D″ the increments can be negative, so the nested intervals need new envelopes, built from the window bounds $[b(n-m) - 2R - (b/4)(n-m), \dots]$. The adaptation is plausible; it is not written and not verified.
  - Lean is one-argument only.

# 3 The exact structure of the one-loop sums

## 3.1 Telescoping [proved+verified, corrected scope]

$S_k := p \to 0$ $F^2$-coefficient of the composed $k$-step gauge-fixed one-loop integral (U1 (1.2)). Then $S_k = \sum_{j=1}^{k} \beta_j(0)$, $S_0 = 0$ (`BETA_FP` Prop. 1.1; `BETA_FP_VERIFY` item 1). Scope: exact for the $p \to 0$ coefficient at fixed $k$, not for $\log Z_k$ at fixed $p$ (the plane wave is off-shell).

## 3.2 Prop. 2.1, anchor-free identity [proved+verified]

Source: `BETA_FP_LEMMA43` Prop. 2.1; `BETA_FP_LEMMA43_VERIFY` items 1, 3–4 (hand rederivation; lattice check with the real CMP 109 averaging at random SU(2) backgrounds, relative difference $10^{-14}$).

Under (G1) = $\{\Delta D = 0,\ \tilde{Q}D = D_c \iota_a^T\}$:

$$\log Z_\Lambda(V) = \big[\log\det X - \tfrac{1}{2}\log\det K\big] - \big[\log\det(D_c^T D_c) - \tfrac{1}{2}\log\det(\Delta_\Lambda(V) + D_c D_c^T)\big].$$

In words: constrained one-loop = fine Wilson one-loop in background Feynman gauge minus the one-loop of the coarse Gaussian theory with kernel $\Delta_\Lambda(V)$. All point-anchor objects cancel; the constant is exactly 0.

- $\tilde{Q}D = D_c \iota_a^T$ is exact at any background (lattice $10^{-15}$). $\Delta D = 0$ is load-bearing (raw Hessian gives $O(1)$ errors).
- At the constant abelian field $V_f$ the background is critical, so $\Delta D = 0$ is exact and Prop. 2.1 holds exactly (`BETA_FP_H_VERIFY` §2, CONFIRMED). At a plane wave it holds up to the defect $\mathfrak{d}_Y$ (`BETA_FP_PW` Lemma 2.3) [proved, unverified].
- Per-cell infinite-volume densities only: on a torus the constant zero modes must be removed. This is input (G2) [formal].

## 3.3 The constant $c$ [proved+verified]

$c = 11N^2/(12\pi^2) = 2N \cdot b_0/(8\pi^2)$, $b_0 = 11N/3$; SU(2): $c = 0.371511$. Normalization $1/g_B^2 = 2N/g^2$ (`R2_VERIFY`; `BETA_FP_LEMMA43_VERIFY` §5). The log coefficient of the fine piece at constant field is $c_2 = c/2$ per $\ln(1/\varphi)$, checked against Savvidy, arXiv:1910.00654 eq. (1.4) (`BETA_FP_H_VERIFY` §4) = input (F) [cited, source checked; lattice = continuum log coefficient is standard, not proved in project files].

$b := c \log L$: 0.25751 ($L = 2$), 0.40815 (3), 0.51502 (4), 0.59793 (5), 0.72293 (7).

## 3.4 The decomposition $S_k = ck \log L + \kappa_W^F - \kappa_k$

- As bookkeeping the split is empty: any $c'$ gives an exact identity if $\kappa_k$ is defined as the remainder (`U1_VERIFY`). Boundedness of a remainder-defined $\kappa_k$ is a restatement, not progress.
- Content comes from defining $\kappa_k$ by the coarse theory alone. `KAPPA_BOUNDED` Prop. 1.1 [proved+verified, conditional on (PI), (F), ($M_j$), V1, (G2)]: with the Wilson level $j = 0$ as reference,

$$S_j = cj \log L - d_j, \qquad \kappa_j = \kappa_W^F + d_j,$$

$$d_j := -\frac{1}{2n_2} \int_0^1 dt \int_{\mathrm{BZ}} \frac{dK}{(2\pi)^4}\ \mathrm{tr}\big[b_{t,2}\,\delta\sigma_j^{(0)} + b_{t,1}\,\delta\sigma_j^{(1)} + b_{t,0}\,\delta\sigma_j^{(2)}\big],$$

where $\delta\sigma_j^{(m)} = \sigma_j^{(m)} - \sigma_0^{(m)}$ are differences of the $f^m$ Taylor symbols of the coarse kernel at level $j$ and level 0, and $b_{t,m}$ are the Moyal inverse coefficients of the interpolated coarse operator. No limit symbols $\sigma_\infty$ are used.

- Checks [numerical, reproduced]: $\rho(1) = \kappa_W^F$ (Lüscher–Weisz (5.7) [16]); $L = 2$: $d_1 = -0.372$, $\kappa_1 = 1.114$; $L = 3$: $d_1 = c\log 3 - \beta_1 = -0.256 \pm 0.01$ (`KAPPA_BOUNDED_VERIFY` §4: fits give $-0.254 \cdots - 0.258$). This checks the definition of $d_j$ through $S_j$; it does not evaluate (1.1) from the symbols independently.

- Cor. 3.1 of `BETA_FP_LEMMA43` ($\mathcal{C}(\Lambda) = c\log\Lambda + \kappa_W^F - \rho(\Lambda)$) is exact as an identity. "The whole $\log\Lambda$ sits in the fine piece" is equivalent to $\rho(\Lambda) = o(\log\Lambda)$. It is not a separate fact (`BETA_FP_LEMMA43_VERIFY` item 2; v1 overclaim removed).

- Given convergence $\beta_j(0) \to \beta_\infty$, Cesàro gives $\beta_\infty = c\log L - \lim\kappa_k/k$. So v1's (H′) "$\kappa_k = o(k)$" is EQUIVALENT to $\beta_\infty = c\log L$, not weaker (`THEOREM2_PROOF_VERIFY` Point 4). Theorem K (§4) gives $\kappa_k$ bounded, hence $\kappa_k = o(k)$ trivially.

### 3.5 Level-uniform symbol facts used below [proved+verified]

Source: `U3_AW.md`, `U3_AW_VERIFY.md` (grades 1–4 CONFIRMED); `U3_REPAIRS_VERIFY`.

- O5/A0: $\Delta_M$ real-analytic on a fixed strip, $|\partial^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, uniform to $K = 0$ (odd $L$, centered blocks). Prop. B: $c_0|\hat{K}|^{-2}P_T' \le P_T' S_M P_T' \le C_1|\hat{K}|^{-2}P_T'$, $c_0 = (2/\pi)^{12}$, $C_1 \le 1363$. Existential constants.

- Prop. E (evenness): $\sigma_j(f;-K) = \sigma_j(f;K)$ in the midpoint trivialization, all $f$, $j$. Tested in $d = 2$ only.

- Theorem W: $\sigma_j^{(1)}(0) = \tau_c = (3ie/2)(E_{10} - E_{01})$ for every $j \ge 0$, from the exact Ward identity $\Delta_j(f)D_f = 0$. Numerically 0.750000, 0.75015, 0.75051 at $j = 0, 1, 2$ vs 0.75 ($d = 2$ proxy).

- Theorem A3: $|\sigma_j^{(1)}(K) - \tau_c| \le C_A|K|^2$, real $K$, uniform in $j$ ($C^0$ only).

- Prop. A2: $\sigma_j^{(m)}$, $m \le 2$, analytic on the same strip at every fixed $j$ (not uniform in $j$ for $m = 1$ derivatives, $m = 2$).

- Lemma M: Maxwell coefficient exactly 1 at every level.

## 4 Bounded window sums: two formally distinct conditional routes

Both routes give the window bound (5.1) of §2.2, $|S_K - S_k - b(K-k)| \le 2R$, with $b = c\log L$. Lemma D″ then gives Theorem 2w (at odd $L > 11$, or with (H0) at this $L$ as a hypothesis, and with L5). Neither route uses convergence of $\beta_j(0)$, a rate, (N), or certified early steps. The two routes have formally distinct named hypotheses. Their mathematical content is not independent: both reduce their uniform-in-level content to the same inequality (H2) on the two-step minimizer map (§5), plus A3-type facts (`BETA_FP_U` §3.5, §4.3 [proved, unverified]; Route K side: `B2_TWOSTEP_VERIFY` grade 3 [proved+verified]).

### 4.1 Route K: Theorem K [proved+verified, conditional on (B2) and the fixed-level inputs]

Source: `KAPPA_BOUNDED.md`; `KAPPA_BOUNDED_VERIFY.md` (items (i)–(v) and (vii) CONFIRMED; (vi) CORRECTED in reasoning, conclusion unchanged).

**Theorem** (Theorem K)**.** *$L$ odd $\geq 3$. Assume (B2) (§5; (B2) $\Leftarrow$ (H2)* [PROVED+VERIFIED]*) and the fixed-level inputs below. Then $\sup_k |d_k| \leq \kappa_* < \infty$, so $\sup_k |\kappa_k| \leq |\kappa_W^F| + \kappa_*$ and $|S_k - ck \log L| \leq \kappa_*$ for all $k$. Window sums satisfy (5.1) with $R = \kappa_*$. (Theorem K is a statement about one-loop quantities. Its use in Lemma D′ for Balaban's $\beta_j(g)$ needs $L$ odd $> 11$ or (H0) at this $L$.)*

*Proof structure.*

1. Prop. 1.1 (§3.4): $S_j = cj \log L - d_j$. [proved+verified, conditional on (PI), (F), ($M_j$), V1, (G2)]
2. Lemma 2.1: the only $K$-derivative of $\sigma^{(1)}$ in $d_j$ enters linearly with $f^0$ coefficients. Integration by parts moves it onto $f^0$ data; boundary terms vanish (periodic up to a common constant conjugation; inner sphere $O(r^2)$). So $\sigma^{(1)}$ is needed only in $C^0$, which Theorem A3 gives. [proved+verified]
3. Prop. 2.2: with (E0) $f^0$ bounds (Prop. B, O5/A0, Lemma M), (E1) Theorem A3 + Theorem W, and (B2), every term of the $d_j$ integrand is $\leq C|K|^{-2}$, uniformly in $j$ and $t$. In $d = 4$ this has two powers of infrared margin. $|d_j| \leq \kappa_* := (C(L)/n_2)(1 + Y_1^2 + C_A(1 + Y_1) + Y_2) \cdot I_2$. [proved+verified, conditional on (B2)]
4. Infrared: the degree $-4$ part of the coarse $f^2$ integrand is the same at every level (Lemma M + Theorem W), so the $\tau_c \log$ cancels exactly in level differences. [proved+verified]

Constants are existential; $\kappa_*$ is not evaluated.

Fixed-level inputs of Route K (needed at each $j$ separately; no uniformity in $j$):

- (PI) probe independence: plane-wave $p^2$ coefficient = constant-field $f^2$ coefficient [argued; per-cell limit for the linear potential not written].
- (F) fine log coefficient $c_2$ at constant field [cited, source checked, §3.3].
- ($M_j$) the formal $f^2$ coefficient $d_j$ equals the true one at fixed $j$. The resolvent step of the first argument was false (Nielsen–Olesen zero at $k_\perp^2 = |ef|$; `BETA_FP_H_VERIFY` §5). `BETA_FP_M` repairs it with a direct $\log|\det|$ bound given spectral bounds (S1)–(S3) [open on the lattice]; Prop. 4.4 there is [argued]; `BETA_FP_M` has no verifier file. The $t$-subdivision of KAPPA §1.3 removes the small-$\varepsilon$ requirement [argued, CONFIRMED as plausible]. Status: [argued].
- V1: the Maxwell coefficient of the per-step determinant equals $\beta_{j+1}(0)$, i.e. order-$g^0$ completeness of CMP 109 (1.3)–(1.5) [open; supported at order $p^2$ by `U3_VERIFY` §6].
- (G2) per-cell infinite-volume densities [formal].
- (H0) Balaban's inductive assumption: $\beta_{j+1}$ smooth on $[0, \gamma]$, "uniformly bounded on this interval together with all derivatives" (CMP 109 pp. 263–264, (1.22)) [Balaban; available at odd $L > 11$ only]. CMP 116 has no theorem statement and no $C^m$-in-$g$ bound in words (`BALABAN_EXCERPTS_CHECK` item 10); the smoothness in $g$ is CMP 109's inductive assumption, asserted by CMP 109 Theorem 3, whose proof CMP 116 completes.
- L5 history-dependent $\beta$, its domain, and its continuity (§2.4) [open].

### 4.2 Route B: Theorem B of BETA_FP_PW [proved+verified, conditional (logic of Theorem B)]

Source: BETA_FP_PW.md §1–§5 (Theorem 3.2 = Theorem B, Theorem 5.1). BETA_FP_PW_VERIFY.md now has a §6 with grades: items 1–4 CONFIRMED (doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B as conditional logic; Lemma 4.1, conditional), items 5 (comparison) and 6 (numerics) CORRECTED. v3 stated that the file had no grade; that is outdated. Tag decision (PLAN.md, 2026-10-28): the logic of Theorem B is [proved+verified, conditional] on its hypotheses, citing BETA_FP_PW_VERIFY §6 items 1–4. Its inputs (U), (DF), (T1) from BETA_FP_U stay [proved, unverified]. The file rederives the Lemma 1.3 arithmetic and the Lemma 2.3 / Prop. 2.2 algebra correctly, confirms the Prop. 2.2 determinant identity on a $4^4$ lattice (4e−13), and notes that identity 3.0 is a tautology ($\Gamma^{ex} := \hat{\Pi} - S$), so all content sits in $(B\theta)/(C\theta)$.

**Theorem** (Theorem B)**.** *$L$ odd $\geq 3$. Assume*

- *(U) there are $p_* > 0$, $C_U$ with $|\beta_{j+1}(p) - \beta_{j+1}(0)| \leq C_U p^2$ for all $j \geq 0$, $0 < p \leq p_*$* [HYPOTHESIS; ARGUED FROM UNIFORM ANALYTICITY IN $p$ OF THE ONE-STEP PLANE-WAVE DATA PLUS ONE-STEP O6]*;*
- *(Bθ) there is $\theta_* \in (0, \min(p_*, 1))$ with $B_* := \sup_k |\Gamma^{ex}(\theta_*; L^k)| < \infty$, where $\Gamma^{ex}$ is the coarse remainder at fixed coarse momentum $\theta_*$* [HYPOTHESIS]*;*
- *Lüscher–Weisz (5.7) [16]* [CITED].

*Then $|S_k - ck \log L| \leq C_B := B_* + c_2 \ln(1/\theta_*^2) + |\kappa_W^F| + \sup_{p \leq \theta_*} |\eta(p)| + C_U \theta_*^2/(L^2 - 1)$, and window sums satisfy (5.1) with $R = C_B$. $\theta_* < 1$ is required so that $c_2 \ln(1/\theta_*^2) \geq 0$; otherwise the term must be replaced by its absolute value (BETA_FP_PW_VERIFY §3).*

*Key step.* Lemma 1.3: $|S(\theta; L^k) - S_k| \leq C_U \theta^2/(L^2 - 1)$ from (U). The massless coarse side is then needed only at a fixed $\theta_* > 0$, where there is no infrared problem. No (W), (PI), (M), IR weights, or spectral theory.

What $(B\theta)$ reduces to (BETA_FP_PW §4) [argued]: Lemma 4.1(i) (trivial-background convergence, from O5, odd $L$) [proved]; vertex bounds at fixed $\theta_*$ (the uniform bounds cited by U3 Theorem 7.1 to Balaban; the fixed-$\theta$ data hypothesis $(D\theta)$ is needed only in its "bounded" form); and boundedness in $k$ of the defect $\mathfrak{d} = \mathfrak{d}_Y + \mathfrak{d}_{gf}$.

- $\mathfrak{d}_Y$ (Prop. 2.1 defect at the non-stationary plane wave): $O(a\Lambda^{-2})$ relative at fixed $\theta$, from zero capacity of point anchors in $d = 4$ [argued; region $|K| \lesssim \Lambda^{-1}$ not written].
- $\mathfrak{d}_{gf}$ (composite gauge vs $D_k^T$ gauge at the plane wave): $O(\theta^2)$ relative; uniformity in $k$ not shown [argued]. Data: 0.953 (direct) vs 0.9595 (recursion) at $L = 2$, $p_0 = 0.2$; difference below the tabulation error, so unresolved.

*Identification.* Route B's $\beta_j(0)$ is the per-level plane-wave increment (the quantity the code computes). It matches Balaban's second-moment $\beta$ up to the expansion point (fine wave vs constrained minimizer), argued to change $S$ by $O(\theta^2)$ relative (SOURCES_D1_VERIFY item 8) [argued]. This identification and the $\mathfrak{d}_{gf}$ bound need expansion-point bounds uniform in $k$. So the (PI)-type content of Route K is not removed in Route B; it moves into a uniform-in-$k$ input (BETA_FP_PW_VERIFY §3).

Three points from BETA_FP_PW_VERIFY §3 (item 5 of its §6 grades them CORRECTED):

1. $\theta_* < 1$ is required (above).
2. The $\beta$ identification (B4) is a uniform-in-$k$ input, not a fixed-level input.

3. (U) is a uniform-in-$j$ analyticity of the level data. It plausibly needs the plane-wave analogue of a uniform $C^1$ bound on $\sigma^{(1)}$: uniform $p$-smoothness of the $a^1$ vertex. That is the same uniform first-order vertex smoothness as Route K's open core ($C1\sigma$).

### 4.3 Route B inputs after BETA_FP_U [proved, unverified; no verifier file]

Source: `BETA_FP_U.md` §2–§4. It reduces the two uniform hypotheses of Theorem B.

- (U) $\Leftarrow$ (T1) + (H2) (Theorem 2.4, odd $L \geq 3$, with A0, A1, Lemma 2.2). (T1): $\sup_j \sup_q |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of Theorem A3 [argued]. (H2) is the inequality of §5 (`BETA_FP_U` writes it as $\tilde{h}_{(2)} < L$).
- ($B\theta$): $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ exactly (Prop. 3.1). $\gamma_{rec}$ is bounded in $k$ given Theorem 2.4 (Prop. 3.2). $\mathfrak{d}_F$ is bounded under (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$ [hypothesis, argued]. `BETA_FP_U` corrects `BETA_FP_PW` §4.3(b): $\mathfrak{d}_F$ is $O(1)$ relative at fixed $\theta$, not $O(\theta^2)$; only boundedness is used.
- Corollary 4.1: given (T1), (H2), (DF), LW (5.7), $|S_k - ck \log L| \leq C_B$, so (5.1) holds with $R = C_B$.
- Theorem 2w by Route B then needs in addition A6 (plane-wave $\beta$ = Balaban's $\beta$ up to the expansion point, uniform in $k$) [argued] and (H0) (Lemma D″ (i)).

Route B hypothesis set in v4: (T1), (H2), (DF), LW (5.7), A6, (H0) (plus L5, (G2), odd $L$, as for Route K). The fixed-level inputs (PI), ($M_j$), V1 of Route K do not appear; (DF) and A6 replace them.

### 4.4 Comparison

| | Route K | Route B |
|---|---|---|
| uniform-in-level input | (B2): $\sup_j \sup_K \|\sigma^{(2)}_j\| < \infty$, $\Leftarrow$ (H2) [proved+verified] | (U), ($B\theta$), and the $\beta$ identification and $\mathfrak{d}_{gf}$ bound, both with $k$-uniform expansion-point bounds; (U) $\Leftarrow$ (T1) + (H2), ($B\theta$) $\Leftarrow$ Theorem 2.4 + (DF) [proved, unverified] |
| single uniform inequality | (H2) | (H2) |
| fixed-level inputs | (PI), (F), ($M_j$), V1, (G2) | LW (5.7), $\mathfrak{d}_Y$, $\theta_* < 1$ |
| uses (W), Theorem A3 | yes | no; `BETA_FP_U` uses (T1), the plane-wave analogue of A3 [argued] |
| verification | verified (conditional) | logic of Theorem B verified (conditional): `BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED [proved+verified, conditional]; inputs from `BETA_FP_U` [proved, unverified], no verifier file |
| common | (H0) at odd $L > 11$, L5, (G2), V1-type identification, Balaban's proofs, SU(2) | |

Shared inputs: (H0), L5, (G2), odd $L$ ($> 11$ for (H0)), and the identification of the computed $\beta_j(0)$ with Balaban's $\beta_j(0)$ (V1; in Route B also the expansion point). The uniform-in-level hypotheses ((B2) vs (U) + ($B\theta$)) are formally distinct as named hypotheses. Their content overlaps:

both rest on (H2) (§5). In Route K, (H2) gives ($C1\sigma$) (Theorem C1) and then (B2) (Theorem B2). In Route B, (H2) gives (U) with (T1), and the vertex bound behind ($B\theta$). So a proof of (H2) serves both routes. One closed route suffices for Theorem 2w.

# 5 The hypothesis (B2) of Route K, and the single uniform inequality (H2)

(B2): $\sup_{j\geq 0}\sup_{K\in \mathrm{BZ}}|\sigma_j^{(2)}(K)| < \infty$. Weaker form that suffices, (B2w): $\sup_j \int_{\mathrm{BZ}}|K|^{-2}|\sigma_j^{(2)}(K)|\,d^4K < \infty$ ($\sigma^{(2)}$ enters $d_j$ only undifferentiated; `KAPPA_BOUNDED` Remark 2.3) [proved+verified: not graded by name; implied by `KAPPA_BOUNDED_VERIFY` §9(iv)].

$\sigma^{(2)}$ obeys the one-step recursion (`U3_AW` (1.2)): $\sigma_j^{(2)} = L^{-2}\mathcal{A}_j\sigma_{j-1}^{(2)} + Q_j[\sigma_{j-1}^{(1)}] + g_j^{(2)}$, with $\mathcal{A}_j s = L^{-2}H_j^{\dagger}(\oplus_l s(k_l))H_j$ and $H_j(K)$ the one-step gauge-fixed minimizer map. Notation: $\rho_1 := \sup_{j,K}\|H_j(K)\|/L$ (one step); $H_{(2),j}(K) := (\oplus_l H_{j-1}(k_l))H_j(K)$ (two steps), $\rho_2 := \sup_{j\geq 2}\sup_{K\ \mathrm{real}}\|H_{(2),j}(K)\|/L^2$.

## 5.1 Proved

- $K=0$ bound [proved+verified]: $\sup_j|\sigma_j^{(2)}(0)| \leq C(|e|C_A + e^2)$. Theorem A3 + scaling give $|\partial^2\sigma_j^{(1)}(0)| \leq 2C_A$ (polarization; `KAPPA_BOUNDED_VERIFY` §7), and the slaving identity (W-2) expresses $\sigma_j^{(2)}(0)$ through these second derivatives.

- Each fixed $j$ [proved+verified]: $\sigma_j^{(2)}$ is analytic on a fixed strip (Prop. A2), hence bounded. The content of (B2) is uniformity in $j$.

- Prop. 4.1 [proved+verified, conditional]: if (h) $h := \sup_{j,K}\|H_j(K)\| < L^2$ and ($C1\sigma$) $\sup_j\sup_K|\partial_K\sigma_j^{(1)}(K)| < \infty$, then (B2) holds. Proof: in the form norm $\pm s \leq N(\Delta+1)$, $N_j \leq (h^2/L^4)N_{j-1} + Z_2$, and $Z_2$ is finite by Theorem A3, ($C1\sigma$), and fixed $f^0$ vertices.

- Why the Lemma A4 mechanism alone fails [proved+verified, reason corrected by `KAPPA_BOUNDED_VERIFY` §7]: the obstruction is the missing uniform $C^1$ bound on $\sigma^{(1)}$, which enters twice: in $Q_j$ (term $L^{-3}\partial_K\sigma_{j-1}^{(1)}$) and in the longitudinal part of $\sigma^{(2)}$ given by (W-2). Theorem A3 is $C^0$ only.

## 5.2 Refuted, and the failed one-step bound

- (HS) $h < L\sqrt{L-1}$ (named (H′) in `U3_AW`) [refuted at $L=3$]. Measured $\max_K\|H_j(K)\|/L =$ 1.652, 1.736, 1.757, 1.760 for $j=1,2,3,\infty$ (`U3_AW_VERIFY` §6); the threshold is $\sqrt{2}=1.414$. Gauge choice cannot help: $H_j$ is already minimal-norm in its gauge orbit.

- One-step sup-norm $C^1$ recursion at $L=3$ [numerical, reproduced; `B2_TWOSTEP_VERIFY` grade 2]. Its factor is $\rho_1^2/L = 0.909$, 1.0046, 1.0289, 1.0325 for $j=1,2,3,\infty$, so it exceeds 1 for $j\geq 2$ ($\rho_1 > \sqrt{3} = 1.7321$). At $K=0$ the factor is $L^{-1}$; the bad region is the face $K_3 = K_4 = \pi$, $K_1 = K_2 \approx 1.86$–$1.92$. This shows that the one-step sup-norm bound fails at $L=3$. It does not show that a $C^1$ bound fails; a weighted norm is not excluded.

### 5.3 (H2) and the two-step theorems [proved+verified, conditional on (H2)]

Source: `B2_TWOSTEP.md` §1–§3; `B2_TWOSTEP_VERIFY.md` grades 1, 3, 4 (CONFIRMED).

$$\text{(H2)} \quad \rho_2 = \sup_{j\geq 2} \sup_{K\in \mathrm{BZ\ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L.$$

(H2) is a plain operator-norm inequality on real $K$. It uses no strip, no $\Delta$-relative weight, and only $n = 2$.

- Two-step map. $\mathcal{A}_j\mathcal{A}_{j-1}s = L^{-4}H^{\dagger}_{(2)}(\oplus s)H_{(2)}$, with $dk_{l,l'}/dK = L^{-2}$ [CONFIRMED, grade 1].
- Theorem C1: (H2) $\Rightarrow$ ($C1\sigma$), with contraction factor $q_1 = \rho_2^2/L^2$ on the derivative. The chain-rule gain is exactly $L^{-2}$ per two steps on real $K$. So the threshold is $\rho_2 < L$, not $\sqrt{L^2-1}$: no strip-Cauchy loss enters. The large constants ($b(L)$ of O6, $\eta_1$, $\eta_2$, $G_1$) enter only the source term, not the contraction factor [CONFIRMED, grade 3].
- Theorem B2: (H2) $\Rightarrow$ (B2), hence (B2w), in plain sup norm, with factor $q_2 = \rho_2^2/L^4$. This contraction needs only $\rho_2 < L^2$, which (H2) implies [CONFIRMED, grade 3]. In one step it is Prop. 4.1's (h).
- (H2) vs (H″). (H2) is the plain operator norm of the same matrix as `U3_AW` (H″) at $n = 2$. It is weaker: threshold $L$ (3 at $L = 3$) against $\sqrt{L^2-1}$ (2.828), real $K$ only, $n = 2$ only [CONFIRMED, grade 4]. (H″) $\Rightarrow$ (B2) stays [argued] and is no longer needed.
- Form of $Q_j$ (G3) [CONFIRMED by read-off of `U3_AW` (1.2)/(3.3) and P10]: $\partial\sigma^{(1)}$ enters linearly. `U3_LEMMA51` (3.3) itself was not re-derived from the Moyal rule by either party.
- Not covered: $C^2$ and $C^3$ bounds on $\sigma^{(1)}$ under (H2) (`B2_TWOSTEP` §5, "look routine", not written, not graded). They are not needed for (B2).
- Submultiplicativity: $\rho_2 \leq \rho_1^2$ always. At $L = 3$ the gain of the two-step map is that this is strict by a factor 1.50 (2.064 vs 3.098) [CONFIRMED].

### 5.4 Numerics for (H2) [numerical; none certified]

Code: `numerics/h2_scan.py` (umklapp Schur pieces, one colour, Gram form; `H2_LARGE_L` §1). Code structure CONFIRMED (`H2_LARGE_L_VERIFY` grade 1): the one-colour reduction is exact at $U = 1$, the $H$ and $\Delta_{out}$ blocks and the Gram identity are correct, the separable Prop. A form is exact. Sign slip in `H2_LARGE_L` §4 ($H = +[\mathfrak{B}^{-1}]_{f,c}$); harmless.

| $L$ | $j$ | $\rho_2$ (scan max) | threshold $L$ | status |
|---|---|---|---|---|
| 3 | 2 | 2.00857 | 3 | [numerical, reproduced] (`B2_TWOSTEP_VERIFY` §4; `H2_LARGE_L_VERIFY` §1) |
| 3 | 3 | 2.05601 | 3 | [numerical, reproduced] (spot value 2.05599 at the argmax) |
| 3 | $\infty$ | 2.0634 (2.06332–2.06377 over $l_{max}$ 2–5) | 3 | [numerical, reproduced]; margin 31%; $q_1 \leq 0.473$, $q_2 \leq 0.053$ |

| | | | | |
|---|---|---|---|---|
| 5 | 2, 3, 4, $\infty$ | 2.098, 2.108, 2.108, 2.108 | 5 | [numerical, reproduced]: independent dense code gives 2.098283 at $j = 2$; the verifier's own search gives 2.107921 at $j = \infty$, same argmax; margin 58% |
| 7 | $\infty$ | $\geq$ 2.1131 (face diagonal only) | 7 | [numerical, run by the verifier] (spot 2.11309) |
| 9 | $\infty$ | $\geq$ 2.1142 (one point, $(2.6, 2.6, \pi, \pi)$) | 9 | [numerical, run by the verifier]; the author's $j = 2$ point gives $\geq 2.1104$ |

- Shape. The argmax lies on the face $K_3 = K_4 = \pi$, on the diagonal $K_1 = K_2$: $\approx 2.22 \to 2.40$ ($L = 3$, $j = 2 \to \infty$), $\approx 2.54$ ($L = 5$). Off-face points are lower: best off-face 1.995 ($L = 3$, $j = 2$), 2.004 ($L = 3$, $j = \infty$), 2.088 ($L = 5$, $j = \infty$). The corner $(\pi, \pi, \pi, \pi)$ stays below the face peak, but the gap shrinks: 0.0070, 0.0028, $\geq 0.0025$, $\geq 0.0023$ ($L = 3, 5, 7, 9$). The corner is the obvious place for a competing maximum at large $L$.

- Level dependence. At $L = 5$, $\rho_2(\infty) - \rho_2(j) = 9.64 \cdot 10^{-3}$, $4.0 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ ($j = 2, 3, 4$); ratios 24.1, 25.0, against $L^2 = 25$. At $L = 3$ the ratios are 7.5, 8.7. Levels $j = 4, \ldots, \infty - 1$ are not sampled at $L = 3$.

- $L$-trend. $j = \infty$: 2.0633, 2.1079, $\geq 2.1131$, $\geq 2.1142$ ($L = 3, 5, 7, 9$). This is consistent with a bounded $\rho_2$ near 2.11–2.14. The rate claim "at least as fast as $L^{-2}$" is CORRECTED (`H2_LARGE_L_VERIFY` grade 5): it used a lower bound at $L = 7$ as if it were the sup. "(H2) gets easier with $L$" is a measured trend for $L \leq 9$, not a result for $L > 11$.

- All values are lower estimates of a sup (grid plus local search).

### 5.5 One-step sufficiency at $L \geq 5$ [arithmetic proved; inputs numerical]

$\|H_{(2),j}(K)\| \leq \max_l \|H_{j-1}(k_l)\| \cdot \|H_j(K)\| \leq L^2 \rho_1^2$. So $\rho_1^2 < L$ implies (H2), and then the one-step $C^1$ recursion of `B2_TWOSTEP` (2.2) contracts with factor $\rho_1^2 / L$.

- $L = 3$: $\rho_1^2 = 3.10 > 3$. The one-step route fails (§5.2); the two-step map is needed.

- $L = 5$: $\rho_1 = 1.860$ ($j = 1$, Wilson fine kernel, dense, measured by the verifier), 1.9565 ($j = 2$), 1.963 ($j = \infty$). So $\rho_1^2 \leq 3.853 < 5$ and $\rho_1^2 / L = 0.771$ [numerical, CONFIRMED as a measurement, uncertified; `H2_LARGE_L_VERIFY` grade 4].

- $L = 7$: $\rho_1 \geq 2.032$ (diagonal only), $\rho_1^2 / L \approx 0.59$ [numerical].

So at $L \geq 5$ the one-step map suffices for Theorem C1, as far as measured. For certification this replaces $L^8$ blocks per $K$ (two-step map) by $L^4$ blocks per $K$ (625× fewer at $L = 5$).

### 5.6 Certification plan and $j_0$ [designed, not done]

`B2_TWOSTEP_VERIFY` grade 6 (CORRECTED in size): certifying (H2) is a computer-assisted proof project, not a check. `H2_LARGE_L` §4 gives the design; `H2_LARGE_L_VERIFY` grades 6–8 check it. Three parts.

(a) Tail $j \geq j_0$. From the resolvent identity, $\rho_2(j) \leq R_\infty + (\rho_1^2/a_*)(\delta_{j-2} + \delta_{j-1})$, where $R_\infty = \rho_2$ at $j = \infty$, $a_* = \inf_K a_j(K)$ with $\|[\mathfrak{B}^{-1}]_{f,f}\| = 1/a_j(K)$ exactly, and $\delta_m = \sup_K \|\Delta_{L^m} - \Delta_\infty\|$ (O5 decay) [derivation CONFIRMED]. Inputs CORRECTED by the verifier: the 2-norm sup of $\delta_m$ is at the corner, $\delta = 179, 92.0, 18.7, 2.39, 0.317$ ($M = 1, 3, 9, 27, 81$), 45–90× the assumed value at small $m$; $a_*(L = 5) = 0.0107$, not 0.05. With these inputs, $j_0 \approx 8$ ($L = 3$) and $\approx 6$ ($L = 5$). With the O6 coercivity constant, $j_0 = 25$, which cannot be used in practice (`B2_TWOSTEP_VERIFY` §5 estimated $j \gtrsim 45$ with $\|\mathfrak{B}^{-1}\| \leq b(L) \sim 10^{19}$).

(b) Levels $2 \leq j < j_0$ and $j = \infty$. A certified sup over the BZ of $\|H_{(2),j}(K)\|$ (at $L \geq 5$: of $\|H_j(K)\|$, §5.5). Plan: interval arithmetic with local mean-value enclosures on boxes in the sorted chamber (1/384 of the BZ), $4 \times 4$ and $5 \times 5$ verified inverses, one $4 \times 4$ $\lambda_{max}$ per $K$; about $2.5 \cdot 10^3$ boxes at half-width 0.05 at $L = 3$; the analytic form of $\Delta$ near $K = 0$, where $\rho_2(0) = 1$. At $j = \infty$ also a certified tail bound for the Prop. A sum ($l_{max}$ truncation; observed effect $4 \cdot 10^{-4}$). There is no explicit global Lipschitz constant today: the A0/O5 strip width is existential, and the constant that can be made explicit is $\sim 10^{40}$. Monotonicity in $j$ is not proved and is not used.

(c) Inputs to certify first: $\|\mathfrak{B}_j(K)^{-1}\| \leq B_{num}$ for all $j$, $K$ (the O6 constant $b(L)$ is unusable), the O5 constant $C_0$ explicitly (or certified $\delta_m$), certified $a_*$, $\rho_1$, $R_\infty$.

Operational fixes for the Windows runs (`H2_LARGE_L_VERIFY` grade 8): atomic JSON writes, pre-copied profile caches, profile build time at $L \geq 11$ not budgeted. Nothing is launched.

### 5.7 The precise remaining inequality

At a fixed odd $L \geq 3$ the single uniform hypothesis is (H2).

- Route K: (H2) ⇒ ($C1\sigma$) (Theorem C1) ⇒ (B2) (Theorem B2) [proved+verified, conditional on (H2)].
- Route B: (H2) + (T1) ⇒ (U), and (H2) ⇒ the vertex bound for $\gamma_{rec}$ in ($B\theta$) (`BETA_FP_U` Theorem 2.4, Prop. 3.2) [proved, unverified].
- Status of (H2): [numerical at $L = 3, 5, 9$; certification open]. At $L = 9$ there is one point only. No value exists at $L > 11$ (Balaban's range), so the numbers are supportive only.
- Kept as alternatives: ($C1\sigma$) + (h) ⇒ (B2) [proved+verified]; (H″) ⇒ (B2) [argued]; (B2w) directly. No per-momentum $d = 4$ constant-field code exists, so (B2) itself has not been scanned numerically (`KAPPA_BOUNDED_VERIFY` §8).

## 6 Conclusion

Theorem 2w ⇐ (Route K inputs) or (Route B inputs). Precisely:

*Route K.* (B2) [open; reduces to (H2) (verified reduction, §5.3); (H2) numerical at $L = 3, 5, 9$, certification open] + (PI), (F), ($M_j$), V1, (G2) [fixed-level] + (H0) [Balaban, odd $L > 11$] + L5 [open] ⇒ Theorem K ($R = \kappa_*$) ⇒ (5.1) ⇒ Lemma D″ ⇒ Theorem 2w. Status of the implication: [proved+verified, conditional].

*Route B.* (U) + ($B\theta$) with $\theta_* < 1$ + LW (5.7) + bounded $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ + identification of $\beta$ (uniform in $k$) + (G2) + (H0) [odd $L > 11$] + L5 ⇒ Theorem B ($R = C_B$) ⇒ (5.1) ⇒ Lemma D″ ⇒ Theorem 2w. Status: Theorem B logic [proved+verified, conditional]; inputs [proved, unverified]. With `BETA_FP_U` (§4.2a): (T1) + (H2) + (DF) + LW (5.7) ⇒ Corollary 4.1 ($R = C_B$); with A6 + (H0) + L5 + (G2) ⇒ Theorem 2w. Status: [proved, unverified, conditional] (the inputs (T1), (DF), A6 and the `BETA_FP_U` reductions are unverified).

*Single uniform hypothesis.* Both routes need (H2). A certified (H2) at one odd $L > 11$ would discharge (B2), the only uniform-in-level hypothesis of Route K (Theorems C1, B2). In Route B it would reduce (U) to (T1); (DF) and A6 would remain.

*Then Balaban's Theorem 1.* Theorem 2w gives (b) and (c′). With $\gamma$ shrunk so that (S) holds, every one of the 19 cross-scale uses in CMP 119 and CMP 122 I–II is covered (§2.3), and the single-scale uses need only $g_k \le \gamma$. So Theorem 1 holds for the bare coupling $g_0(K, g)$, for every $K$ and every $g \in ]0, g_1]$, without Balaban's coupling assumption. This needs L5 in the sharpened form of §2.4: $\beta$ must exist on every history the shooting visits. Status: [proved, unverified (sweep), conditional on Route K or Route B inputs, on L5, and on Balaban's published proofs].

*Order of constants.* Fix $\beta_0, p_0, p_1, r, \nu, \kappa_0$ (Balaban). Fix $L$ (odd, $> 11$), then $R$ from Route K or B, then $b = c \log L$, then $\gamma \le \min(\gamma_0, \gamma_{Bal}, b/(4C_1), \gamma_*)$ with $\gamma^{-2} \ge c_{sw} \cdot 2R$. Then $g_1 = (\gamma^{-2} + 2R)^{-1/2}$. Balaban's phrase "for a sufficiently small positive $g$" fits $g \le g_1$.

*What this does not give.*

- Literal (0.31): only for windows of length $\ge 8R/b$. Literal (0.31) at every $K$ needs (L1) or (L2) of §1.4, which need individual increments, not window sums. Not needed for Theorem 1.
- The value $\lim \beta_j(0)$ and any rate: §7.
- Even $L$, or odd $L \le 11$ without (H0) as an extra hypothesis (Balaban assumes $L$ odd $> 11$ throughout CMP 109); SU($N$) with $N > 2$.
- Monotone couplings: $g_m$ can exceed $g_n$ for $m < n$ by the bounded amount $u_m \ge u_n - D'$. §2.3 shows Balaban's proof tolerates this under (S); a line-by-line read was not done.
- A continuum limit (`ASSEMBLY.md` treats the assembly separately; not claimed here).

# 7 The convergence route: what it adds, and its status

Not needed for Theorem 1. It adds: (i) existence of $\beta_\infty = \lim \beta_j(0)$; (ii) its value $c \log L$; (iii) a rate; (iv) eventual positivity, hence literal (0.31) for $K \ge K_0$ (Lemma D′ (d)); (v) with certified early steps, literal (0.31) at every $K$ (§1.4 (L2)). It also implies (B2), since convergence of $\sigma_j^{(2)}$ implies boundedness. So it needs strictly more than Route K.

## 7.1 Components and status

- U3 Theorem 4.1 with O5: $|\partial_K^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, $|\alpha| \le 2$, uniform to $K = 0$, odd $L$ [proved+verified] (`U3_VERIFY`, `U3_REPAIRS_VERIFY`). Numerically $|\Delta_M - \Delta_\infty| = 1.007, 0.129, 0.0147$ at $M = 3, 9, 27$.
- O6: $\|\mathfrak{B}_j(K)^{-1}\| \le b(L)$ for all $j$, $K$, one-step constraint, trivial background [proved+verified]. Crude constant ($b(3) \approx 7.5 \cdot 10^{33}$); measured corner values 103.5, 176.7, 193.0, 195.3 ($j = 0, 1, 2, \infty$, $L = 3$) use a surrogate $\Lambda$ (`U3_REPAIRS_VERIFY` §5).
- Theorem 6.1: $\beta_{j+1}(0) = \Phi(\mathcal{D}_j)$ is Lipschitz in the level data [proved+verified, conditional on V1]. The norm is the Taylor-data norm (`U3_VERIFY` P8 correction).
- Lemma 5.1 (convergence of $\sigma_j^{(1)}$, $\sigma_j^{(2)}$ in $C^2$, $|\alpha| \le 2$):
  - neutral sector, rate $L^{-4j}$ [proved+verified for the value bound; the $K$-derivative paragraph is proved, unverified];

  - (W) [proved+verified] (`U3_AW` Theorem W); it removes the marginal direction of the $f^1$ recursion;
  - (A) qualitative and Theorem A3 [proved+verified]; (A) in full (strip/$C^3$ for $m = 1$, all of $m = 2$) $\Leftarrow$ (H″) [open]; the route via (HS) is [refuted] at $L = 3$;
  - $\sigma^{(1)}$ convergence at rate $jL^{-2j}$ [sketch in `U3_AW` §4.7, unverified]; $\sigma^{(2)}$ convergence [open, needs (A) in full];
  - the single-block-constraint version of Lemma 5.1 is false for $m = 2$ in $d = 4$ [proved+verified]; Balaban's composed constraint is not affected.

- Prop. 3.6: $|\beta_{j+1}(0) - \beta_\infty| \le C(1+j)^2 L^{-2j}$ [proved, unverified as an assembly; conditional on V1, (A) in full]. (v1 wrote $Cj^2L^{-2j}$, which is 0 at $j = 0$; corrected.) The identification of $\sigma$-Taylor data with the data of (6.1) is cited from `U3_VERIFY` §7, not re-shown. (N) (clean rate $L^{-2j}$) is not needed and is judged plausibly false.

- Value of the limit. Given convergence, Theorem K or Theorem B gives $\beta_\infty = c \log L$ at once: $S_k/k \to \beta_\infty$ by Cesàro and $S_k/k = c \log L + O(1/k)$ [proved+verified as an implication: `THEOREM2_PROOF_VERIFY` Point 4; `BETA_FP_PW` Theorem 3.2(iii)]. Older routes to the value: `BETA_FP_H` Theorem 3.1′ with (H) restated at the constant field, which given (W) reduces to Lemma 5.1 (1.2) alone (`BETA_FP_H_VERIFY` items (1)–(4)), plus (PI), (F), (M) [conditional; (M) argued]; and `BETA_FP_PW` Theorem 3.1 (limit form) given (U), ($C\theta$), LW (5.7) [proved, unverified]. The original (H) with IR weights (`BETA_FP_LEMMA43` (4.2)) was graded UNSUPPORTED and is not used.

- Rate of $\mathcal{C}(\Lambda)$: $O(\Lambda^{-2} \log \Lambda)$ given Lemma 5.1 with rates, via `U3_AW` §3.4(b) slaving [argued, `BETA_FP_H_VERIFY` §5].

### 7.2 Early steps and $j_*$

Under Prop. 3.6 and $\beta_\infty = c \log L$, $\beta_{j+1}(0) \ge (c \log L)/2$ for $j \ge j_* := \min\{J : C(1+j)^2 L^{-2j} \le (c \log L)/2 \text{ for } j \ge J\}$. Heuristic size [argued]: $C \sim b^3$ gives $j_* \approx 110$ ($L = 3$), $\approx 70$ ($L = 13$); $C \sim b^6$ (one-step inverses inside $C_{data}$) gives $\approx 218$, $\approx 138$ (`THEOREM2_PROOF_VERIFY` Point 5(e)). With the measured $\|\mathfrak{B}^{-1}\| \approx 195$: $j_* \approx 10$ ($L = 3$), $\approx 4$ ($L = 13$). Certifying $\beta_1(0), \dots, \beta_{j_*}(0) > 0$ needs interval arithmetic at odd $L \ge 13$ (blocks $\approx 3.4 \cdot 10^5$ rows per momentum by dense methods; the umklapp code reduces this to $12 \times 12$ blocks at $L^4$ momenta). It is needed only for (L2), not for Theorem 1.

## 8 Numerics

SU(2), infinite volume, Balaban's normalized-trace units. None of this is certified. Only odd $L$ is inside the proofs; $L = 2, 4$ use corner blocks and are outside them.

### 8.1 First increment $\beta_1(0)$ (`NUMERICS` §6–§7; `UMK_BETA1.md`)

| $L$ | $\beta_1(0)$ | how | $b = c \log L$ | $\beta_1 - b$ | status |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| 2 | 0.630 | $N_g = 12$, $p \to 0$ fit | 0.2575 | +0.372 | [numerical, reproduced] (KAPPA_BOUNDED_VERIFY §4: 0.6296) |
| 3 | 0.664 | $N_g = 8$, $p \to 0$ fit (0.6625–0.6661) | 0.4082 | +0.256 | [numerical, reproduced]; $N_g$ 6 $\to$ 8 shift $\approx 0.015$ |
| 4 | 0.757 | $N_g = 6$, single $p = 0.2$; $O((Lp)^2)$ bias $\approx 0.01$ | 0.5150 | +0.242 | [numerical] |
| 5 | 0.749 ($N_g = 12$, $p = 0.1$); fit 0.743 | umklapp code | 0.5979 | +0.145 | [numerical] |
| 7 | 0.803 | umklapp code, $N_g = 8$, $Lp = 0.5$ | 0.7229 | +0.080 | [numerical] |
| 9 | 0.833 ($Lp = 0.5$); 0.846 ($Lp = 0.9$), 0.851 ($Lp = 1.0$) | umklapp code, Windows, $N_g = 8$ | 0.8163 | $\approx +0.02$ | [numerical] |
| 11 | 0.850 ($Lp = 0.5$); 0.868 ($Lp = 1.0$) | umklapp code, Windows, $N_g = 8$; $N_g = 8$ grid adequacy at $L = 11$ unchecked | 0.8908 | $\approx -0.04$ | [numerical] |

The first-step constant $\beta_1 - c \log L$ is +0.37, +0.26, +0.24, +0.15, +0.08, +0.02, −0.04 for $L = 2, 3, 4, 5, 7, 9, 11$. It decreases with $L$ and changes sign between $L = 9$ and $L = 11$. $\beta_1(0)$ itself stays positive at every $L$ computed. The umklapp code was validated against `ym_momentum` to $10^{-12}$ per $K$ by its author; no independent verifier for the $\beta_1$ values at $L \geq 5$. The queue continues to $L = 15$ on Windows. No value at $L > 11$ exists yet.

### 8.2 Later increments ($L = 2$ only)

| quantity | value | status |
|---|---|---|
| $\beta_1 + \beta_2$ (composed, $N_g = 4/6/8$) | 1.148 / 0.967 / 0.953 | [numerical]; single $p = 0.2$, $Lp$ bias label |
| per-level $\beta_2(0)$ | $\approx 0.32$ (0.953 − 0.630). Per-level recursion 0.317–0.341 ($N$ = 8–12, single $p = 0.2$). Symmetric Nyquist: 0.33334 ($N = 8$, $p_1 = 0.2$), 0.31679 ($N = 12$, $p_1 = 0.2$), 0.32283 ($N = 12$, $p_1 = 0.4$) (BETA_FP_VERIFY §7; BETA_FP_PW_VERIFY §5–§6). Level-1 values carry $N$-aliasing of about ±5%. | [numerical]; per-level recursion reproduced in BETA_FP_VERIFY §7 |
| $\beta_3(0)$ | 0.40 ($N = 8$), 0.286 ($N = 12$), 0.261 (verifier, symmetric Nyquist, $N = 8$) | unconverged; not usable |
| abelian sector $S_k^{ab}$ | $(1 - L^{-4k})/4$, matched to all digits | [proved+verified]; [Lean] counting part (AbelianCount.lean) |

| | | |
|---|---|---|
| (U) test | $\lvert\beta_{j+1}(p) - \beta_{j+1}(0)\rvert \le 0.2p^2$ at every computed point ($L = 2$ levels 0–1, $L = 3$ level 0, $p \le 0.8$); uniformity in $j$ not testable with two levels | [numerical, run by the verifier] (`BETA_FP_PW_VERIFY` §6) |

All computed increments are positive.

### 8.3 Quantities of the Route K proof

| quantity | value | status |
|---|---|---|
| $d_1 = b - \beta_1$, $L = 3$ | $-0.256 \pm 0.01$ (fits $-0.254 \cdots - 0.258$) | [numerical, reproduced] |
| $d_1$, $\kappa_1$, $L = 2$ | $-0.372$, $1.114$ | [numerical, reproduced] |
| $d_2$, $d_3$, $L = 2$ | $\approx -0.44$, $\approx -0.45$ ($d_3$ uses unconverged $\beta_3$) | [numerical]; bounded-looking, not a test |
| $\rho_1 = \max_K \lVert H_j(K)\rVert/L$, $L = 3$, $j = 1, 2, 3, \infty$ | 1.652, 1.736, 1.757, 1.760 | [numerical, run by the verifier, `U3_AW_VERIFY` §6]; refutes (HS); $h^2/L^4 \approx 0.34 < 1$; one-step $C^1$ factor $\rho_1^2/L > 1$ for $j \ge 2$ |
| $\rho_1$, $L = 5$, $j = 1, 2, \infty$ | 1.860, 1.9565, 1.963; $\rho_1^2/L = 0.771$ | [numerical]; $j = 1$ and the bound measured by the verifier (`H2_LARGE_L_VERIFY` §6) |
| $\rho_2$ ((H2)), $L = 3, 5, 7, 9$ | see §5.4: 2.0634, 2.108, $\ge$ 2.1131, $\ge$ 2.1142 at $j = \infty$ against thresholds 3, 5, 7, 9 | [numerical, reproduced] at $L = 3, 5$; [numerical, run by the verifier] at $L = 7, 9$ |
| $a_*$, $L = 5$ | 0.0107 (binding mode near $K = 0$, gauge/ghost sector) | [numerical, run by the verifier] (`H2_LARGE_L_VERIFY` §6) |
| (H″), $n = 2$, $L = 3$ | $2.056 < 2.828$ | [numerical, run by the verifier], coarse grid, one level pair |
| $\sigma_j^{(1)}(0)$ $J$-part, $j = 0, 1, 2$ ($d = 2$ proxy) | 0.750000, 0.75015, 0.75051 vs 0.75 | [numerical, run by the verifier] supports Theorem W |
| $\sigma_j^{(2)}(K)$ scan | not run: no $d = 4$ constant-field per-momentum code | — |

## 9 Ledger (tags match the verifier files)

**[Lean]** (no sorry; axioms propext, Classical.choice, Quot.sound)

- Lemma D′ (a)–(d) (`YmLean.lemma_DPrime`), Step 8 (`lemma_DPrime_step8`), Lemma D, Lemma E ($Cg^2$ form), Lemma D+E (`lean/YmLean/LemmaDPrime.lean`, `LemmaDE.lean`). One-argument $\beta$ only.
- Lemma D″ (`YmLean.lemma_DDoublePrime`, `lean/YmLean/LemmaDDoublePrime.lean`, 146 lines): (a″) bracket $(3b/4)(K - k) - 2R \le T(k, K) \le (5b/4)(K - k) + 2R$; (b) shooting for $g \in ]0, g_1]$ with the clamp inactive; (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, for every trajectory with couplings in $]0, \gamma]$; (d) for $n - m \ge 8R/b$, lower constant $b/2$. Hypotheses

weaker than the note (continuity, any admissible $\gamma$, $R \geq 0$ derived). One-argument $\beta$ only; L5 not formalized.

- The Lean folder also has Telescoping, DetIdentities, AbelianCount, and LemmaH files (`lean/README.md`); this chapter uses only the AbelianCount counting part, as a cross-check in §8.2.

**[proved+verified]**

- Lemma D′ (a)–(d), $K_0$, Remark 4 (`LEMMA_DPRIME_VERIFY` grades 1, 2, 5).
- Lemma D (corrected interval), Lemma E ($Cg^2$ form), nested-interval variant of Lemma D (`LEMMAS_DE_VERIFY`). Lemma E $C^1$ form: trivial variant, same proof.
- Lemma D″ (tail-sum form, no $J$) (`KAPPA_BOUNDED_VERIFY` §6).
- Telescoping $S_k = \sum \beta_j(0)$ for the $p \to 0$ coefficient (`BETA_FP_VERIFY` item 1, corrected scope).
- Prop. 2.1 anchor-free identity; exact at the constant field (`BETA_FP_LEMMA43_VERIFY` item 1; `BETA_FP_H_VERIFY` item 1).
- $c = 11N^2/(12\pi^2)$, normalization chain; (F) source (Savvidy eq. (1.4)) (`BETA_FP_LEMMA43_VERIFY` item 2; `BETA_FP_H_VERIFY` §4).
- KAPPA Prop. 1.1 as an identity given its fixed-level inputs; Lemma 2.1 (integration by parts); Prop. 2.2; level-independence of the degree $-4$ part; Theorem K and Theorem 2w as conditional reductions (`KAPPA_BOUNDED_VERIFY` (i)–(v)).
- $\sup_j |\sigma_j^{(2)}(0)| < \infty$; Prop. 4.1 algebra (`KAPPA_BOUNDED_VERIFY` (vi), reason for A4 failure corrected: obstruction = $\partial\sigma^{(1)}$ in $Q_j$ and in (W-2), §7 there).
- `B2_TWOSTEP` two-step recursion; Theorem C1 ((H2) $\Rightarrow$ ($C1\sigma$), factor $\rho_2^2/L^2$) and Theorem B2 ((H2) $\Rightarrow$ (B2), factor $\rho_2^2/L^4$), conditional on (H2); threshold $\rho_2 < L$ (no strip loss); (H2) weaker than (H″) at $n = 2$ (`B2_TWOSTEP_VERIFY` grades 1, 3, 4). Form of $Q_j$ by read-off (grade 8; `U3_LEMMA51` (3.3) not re-derived).
- `h2_scan` code structure, one-colour reduction, $H$ and $\Delta_{out}$ blocks, Gram identity, separable Prop. A form (`H2_LARGE_L_VERIFY` grade 1); tail-bound derivation (4.1)–(4.3) (grade 6, inputs corrected).
- Logic of Theorem B and Lemma 1.3 of `BETA_FP_PW`, as conditional implications (`BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED: doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B; Lemma 4.1, conditional). Its inputs (U), (DF), (T1) are listed under [proved, unverified] and [argued].
- Prop. E evenness, Theorem W0/W, Theorem A3, Lemma A4, A0, Prop. A2 (`U3_AW_VERIFY` grades 1–4).
- U3 Theorem 4.1 + O5, Prop. A, Prop. B, O6, Theorem 6.1 (given V1; Taylor-data norm), Lemma 5.1 neutral value bound, single-block $m = 2$ failure in $d = 4$, one-step recursion, R2 midpoint identity (`U3_VERIFY`, `U3_REPAIRS_VERIFY`, `U3_LEMMA51_VERIFY`, `U3_LEMMA51_R23_VERIFY`).
- Cesàro: given convergence, $\kappa_k = o(k) \Leftrightarrow \beta_\infty = c \log L$ (equivalence, not a weaker form) (`THEOREM2_PROOF_VERIFY` Point 4).

**[proved, unverified]**

- `BETA_FP_U`, no verifier file: Theorem 2.4 ((U) $\Leftarrow$ (T1) + (H2)), Prop. 3.1 ($\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$), Prop. 3.2 ($\gamma_{rec}$ bounded), Corollary 4.1, Theorem 2w by the plane-wave route given (T1), (H2), (DF), LW (5.7), A6, (H0); the argued step "(H2) closes ($C1\sigma$)" of §3.5 there is superseded by `B2_TWOSTEP` Theorem C1 [proved+verified].
- `LEMMA_DPRIME_SWEEP`: 19 cross-scale uses covered by (b) + (c′) + (S) (4 of 19 table rows, 11, 12, 16, 18, verified on page images by `LEMMA_DPRIME_VERIFY` comparisons A–E).
- Step "Theorem 2w $\Rightarrow$ Balaban's Theorem 1" (depends on the sweep and on L5 in the sharpened form of §2.4).
- Prop. 3.6 assembly with $C(1+j)^2 L^{-2j}$; neutral-sector $K$-derivative paragraph; $\sigma^{(1)}$ convergence sketch $jL^{-2j}$.
- Necessity of individual positivity for literal (0.31) at every $K$ (elementary).
- `BETA_FP_M` Prop. 4.3 ($\log|\det|$ bound in the N–O window, given (S1)–(S3)); no verifier file.
- One-step sufficiency $\rho_1^2 < L \Rightarrow$ (H2) (submultiplicativity; arithmetic, `H2_LARGE_L` §3).

**[argued]**

- (PI); ($M_j$) (Prop. 4.4 of `BETA_FP_M`); (U) as a bare hypothesis (now reduced, see above); (T1) (plane-wave form of A3, `BETA_FP_U` §2.3); (DF) (`BETA_FP_U` §3.4); A6 = plane-wave vs constrained-minimizer expansion point (uniform in $k$); $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ bounds; (H″) $\Rightarrow$ (A) in full $\Rightarrow$ (B2) (`U3_AW` §4.7 sketch); $C^2$, $C^3$ of $\sigma^{(1)}$ under (H2) (not written); $j_*$ size; the rate $O(\Lambda^{-2}\log\Lambda)$.

**[open]**

- (H2) (certification; numerical at $L = 3, 5, 9$). (B2) is reduced to it [verified reduction]. (T1), (DF), A6 for Route B. V1. (S1)–(S3). L5 (domain of $\beta$; nested-interval D″ [not written]). $K$-independence of $\beta_j(g)$. (A) in full. See §10.

**[corrected by verifiers since v3]**

- `B2_TWOSTEP` G1: the certification of (H2) is a computer-assisted proof project, not a check (`B2_TWOSTEP_VERIFY` grade 6).
- `H2_LARGE_L`: $L$-trend rate claim (grade 5); $j_0$ inputs, $\delta_m$ 45–90× larger, $a_*(L=5) = 0.0107$ (grade 6), giving $j_0 \approx 8$ ($L=3$), $\approx 6$ ($L=5$).
- `BETA_FP_PW`: comparison and numerics (`BETA_FP_PW_VERIFY` grades 5, 6).

**[refuted]**

- (HS) $\sup\|H_j\| < L\sqrt{L-1}$ at $L = 3$.
- Not a refuted statement, but a failed method: the one-step sup-norm $C^1$ recursion at $L = 3$ (factor 1.005–1.033 for $j \geq 2$; §5.2).

- `U3_LEMMA51` Theorem 4.1 for $m = 2$ with a single block-$M$ constraint in $d = 4$.

- v1 claims withdrawn: "every early step must be certified for Theorem 1" (false: Lemma D′/D″); "(H′) is weaker than $\beta_\infty = c \log L$" (it is equivalent); "the whole $\log \Lambda$ sits in the fine piece" as a fact (it is equivalent to the open statement); "one-loop parts hold for SU($N$)" (Lemma 5.1's charged/neutral split is the SU(2) Cartan split); (3.2) with $Cj^2L^{-2j}$ (0 at $j = 0$); the (M) resolvent bound $\|\mathcal{K}_t^{-1}\| \le C/|f|$ (Nielsen–Olesen zero); (H) with IR weights as stated (UNSUPPORTED).

**[Balaban]**

- (H0) smoothness and uniform bounds of $\beta_{j+1}$ in $g$ (CMP 109 (1.22), pp. 263–264), at odd $L > 11$ only; Theorem 1 and the inductive bounds of CMP 109, 116, 119, 122 and the propagator papers CMP 95, 96, 98, 99, 102. Uniform level bounds partly cited from CMP 95/96/99 (O6 replaced the main citation).

**[verified quote]**

- Theorem 2 and Theorem 1 wording, "separate paper", p. 298 history sentence, (0.1) with $1/g_k^2$ and $\chi_k$, (1.3)–(1.5) (print misprint $E^{(j+k)}$ read as $E^{(j+1)}$), (2.6)–(2.9) of CMP 119, items A–E of CMP 122 (`BALABAN_EXCERPTS_CHECK`; `LEMMA_DPRIME_VERIFY`; `LEMMA_DPRIME_SWEEP` page images). `BALABAN_EXCERPTS_CHECK` items 10 (CMP 116 has no $C^m$-in-$g$ statement) and 11 (unchecked readings) are unchanged since v3.

**[numerical]** see §5.4 and §8.

## 10 Open list (complete for Theorem 2w ⇒ Theorem 1)

Single uniform hypothesis (both routes):

H. (H2) $\rho_2 = \sup_{j\ge 2} \sup_{K \text{ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L$ [numerical at $L = 3, 5, 9$; certification open]. It replaces $(C1\sigma)$ + (h) and (H″) of v3 as the one uniform-in-level inequality. Route K: (H2) ⇒ $(C1\sigma)$ ⇒ (B2) [proved+verified, `B2_TWOSTEP_VERIFY` grade 3]. Route B: (H2) + (T1) ⇒ (U) [proved, unverified, `BETA_FP_U` Theorem 2.4]. Values ($j = \infty$, uncertified lower estimates of the sup): 2.0634 ($L = 3$, threshold 3, margin 31%), 2.108 ($L = 5$, threshold 5, margin 58%), $\ge$ 2.1131 ($L = 7$), $\ge$ 2.1142 ($L = 9$, one point). At $L \ge 5$ the one-step condition $\rho_1^2 < L$ suffices ($\rho_1^2/L = 0.771$ at $L = 5$). Certification (§5.6): a computer-assisted proof project with tail level $j_0 \approx 8$ ($L = 3$), $\approx 6$ ($L = 5$), certified $\|\mathfrak{B}^{-1}\|$, an explicit O5 constant, and interval arithmetic over the BZ; no explicit Lipschitz constant exists today. No value exists at $L > 11$.

Route K hypothesis set (all must hold):

1. (B2) $\sup_j \sup_K |\sigma_j^{(2)}| < \infty$ (or (B2w)). Reduced to H [verified reduction]. Older reductions kept: $(C1\sigma)$ + (h) [proved+verified]; (H″) [argued]. (HS) is refuted.

2. (PI) probe independence, at each fixed level [argued].

3. (F) lattice = continuum log coefficient at constant field [cited source checked; lattice transfer standard, not proved here].

4. ($M_j$) formal = true $f^2$ coefficient at each fixed level: (S1) $\mathcal{K}_t(f) \geq -C_1|f|$, (S2) density of states $\leq C_2|f|$ near 0, (S3) relative bound of the level difference, all on the lattice [open]; Prop. 4.4 [argued].

5. V1 order-$g^0$ completeness of CMP 109 (1.3)–(1.5): $\beta_{j+1}(0)$ is the Maxwell coefficient of the per-step Gaussian determinant [open; supported at order $p^2$].

6. (G2) per-cell infinite-volume densities [formal].

Route B hypothesis set (alternative to 1–5; per `BETA_FP_U` §4; all [proved, unverified] as implications, no verifier file for `BETA_FP_U`):

B1. (T1) $\sup_j \sup_{q \text{ real}} |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of Theorem A3 [argued; the Landau-to-symmetric gauge conversion and Steps 2–5 of A3 for (2.1) are not written].

B2′. (H2) = item H.

B3. (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$, the gauge-fixing difference at the plane wave [hypothesis, argued]. With the exact split $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ (`BETA_FP_U` Prop. 3.1) there is no $Y$-defect; all non-stationarity sits in $\mathfrak{d}_F$. This replaces v3's $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ items for this route.

B4. Lüscher–Weisz (5.7) [cited].

B5. A6: the plane-wave increment equals Balaban's $\beta$ up to the expansion point, uniformly in $k$ [argued]. A uniform-in-$k$ input, of the same kind as (DF).

B6. (H0) = item 7 (Lemma D″ (i)).

B7. $\theta_* < 1$, so that $c_2 \ln(1/\theta_*^2) \geq 0$ in $C_B$ [trivial condition on the choice of $\theta_*$].

B8. Verification: `BETA_FP_U` has no verifier file. `BETA_FP_PW_VERIFY` §6 items 1–4 grade the logic of Theorem B [proved+verified, conditional] (tag decision, `PLAN.md`); the inputs (U), (DF), (T1) stay [proved, unverified].

Route B also needs 6 ((G2)) and the common items 8–13. It does not need (PI), ($M_j$), V1; it needs (DF) and A6 instead.

Common to both routes:

7. (H0) Balaban's smoothness in $g$ with uniform bounds, $|\beta_j(g) - \beta_j(0)| \leq C_1 g$ uniform in $j$ and history [Balaban; CMP 109 inductive assumption at odd $L > 11$; no explicit proof of the $g$-derivative bound located in CMP 116].

8. L5 history dependence [open]. (a) The clamp proof of Lemmas D′/D″ is valid only if $\beta_{k+1}(g_0, \ldots, g_k)$ is defined and continuous on the whole cube $]0, \gamma]^{k+1}$; Balaban does not supply this. (b) Balaban's $\beta_{k+1}$ exists only on histories where his induction has run; that domain must carry (c′) + (S), not only $g_i \in ]0, \gamma]$. Clamped histories can violate the growth side of (c′). (c) True trajectories with all $u_i \geq \gamma^{-2}$ inherit (c′) from (a″). A nested-interval variant of Lemma D″ on these trajectories would avoid the gap [not written]; the variant is written only for Lemma D (`LEMMAS_DE_VERIFY` §1.5). Lean covers one-argument $\beta$ only (Lemmas D′, D″, D, E).

9. $K$-independence of $\beta_j(g)$ [open]. The shooting needs $\beta_j(g)$, $\gamma$, and the constants independent of $K = -\log_L \varepsilon$. After rescaling, each step is the same unit-lattice block step, so $\beta_j$ is $K$-independent only in the infinite-volume limit; a finite volume brings $K$-dependence back. This infinite-volume identification is not proved (`LEMMAS_DE_VERIFY` §2 item 2; `THEOREM2_PROOF_VERIFY` Point 7). It is used implicitly through (G2) and (H0).

10. Balaban readings: the sweep (§2.3) is grep plus page images, not line-by-line, with 4 of 19 rows checked by a second reader; "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) not located; $\beta_0$ must be fixed before $\gamma$, while Balaban says $\beta_0$ "can be chosen arbitrarily small, if $g$ is sufficiently small" (not checked line by line); `BALABAN_EXCERPTS_CHECK` item 11 lists unchecked readings (Π (4.37)/(5.1), CMP 119 (2.42), (2.12)/(2.23), the "$L$ odd" page, Sect. 3 estimates, 1989 bound lists); Theorem 1 itself rests on Balaban's published proofs.

11. Odd $L$. All one-loop proofs need odd $L \geq 3$ (centered blocks: U3 Lemma 1.2, O5/Prop. A, evenness, Lemma 4.1 of LEMMA43). Balaban assumes $L$ odd $> 11$ throughout CMP 109, so (H0) and $\beta_j(g)$ at $g > 0$ exist only there. Theorem 2w is stated for odd $L > 11$, or with (H0) at this $L$ as a hypothesis. Numerics: $\beta_1(0)$ up to $L = 11$ (§8.1); (H2) at $L = 3, 5$ (scans), 7 (face diagonal), 9 (one point). All are outside Balaban's range ($L > 11$), so they are supportive only. The Windows queue continues to $L = 15$.

12. SU(2) only. The charged/neutral split of Lemma 5.1 and (W) are SU(2) statements; $c = 11N^2/(12\pi^2)$ is general but the chain is not checked for $N > 2$.

13. Lean formalization of the history form of Lemmas D′/D″ [open]. Lemma D″ in the one-argument form is done (§2.2).

Not needed for Theorem 1, open for the stronger statements: (A) in full and Lemma 5.1 for $m = 2$ (convergence, rate); $C^2$, $C^3$ of $\sigma^{(1)}$ under (H2) (not written); certified $\beta_1(0), \ldots, \beta_{j_*}(0) > 0$ at odd $L > 11$ (literal (0.31) at every $K$); explicit $j_*$; (N).

**Hypothesis table** (v3 table from `THEOREM2_PROOF_v2_VERIFY` §K, updated to v4 status).

| hypothesis | introduced | verified / checked in | status | §10 item |
|---|---|---|---|---|
| (H2) | B2_TWOSTEP §2; BETA_FP_U §2.3 | B2_TWOSTEP_VERIFY §4; H2_LARGE_L_VERIFY §1–§10 | numerical at $L = 3, 5, 9$; certification open | H |
| (H2) ⇒ ($C1\sigma$) ⇒ (B2) (Theorems C1, B2) | B2_TWOSTEP §2–§3 | B2_TWOSTEP_VERIFY grades 1, 3, 4 | proved+verified, conditional on (H2) | 1, H |
| (H2) + (T1) ⇒ (U) (Theorem 2.4) | BETA_FP_U §2.4 | none | proved, unverified | B1, B2′ |
| $\rho_1^2 < L$ ⇒ (H2) (one step, $L \geq 5$) | H2_LARGE_L §3 | H2_LARGE_L_VERIFY grade 4 (measurement) | arithmetic; $\rho_1^2/L = 0.771$ at $L = 5$, numerical | H |

| | | | | |
|---|---|---|---|---|
| certification tail $j_0$ | H2_LARGE_L §4.1 | H2_LARGE_L_VERIFY grade 6 | derivation confirmed; $j_0 \approx 8$ ($L = 3$), $\approx 6$ ($L = 5$), corrected | H |
| (B2) / (B2w) | KAPPA §2, Rem. 2.3 | KAPPA_VERIFY §7–§9 | open; reduced to (H2) | 1 |
| $(C1\sigma)$ + (h) $\Rightarrow$ (B2) | KAPPA Prop. 4.1 | KAPPA_VERIFY §7 | implication verified; $(C1\sigma) \Leftarrow$ (H2); (h) numerical $L = 3, 5$ | 1 |
| (H″) $\Rightarrow$ (A) $\Rightarrow$ (B2) | U3_AW §4.7 | none (U3_AW_VERIFY grade 5 covers (H′) only) | implication argued; not needed after (H2) | 1 |
| (HS) | U3_AW (H′) | U3_AW_VERIFY §6 | refuted $L = 3$ | §5.2 |
| (PI) | BETA_FP_H §3.1 | H_VERIFY §5 | argued | 2 |
| (F) | BETA_FP_LEMMA4.3 | H_VERIFY §4 | cited, source checked | 3 |
| $(M_j)$: (S1)–(S3), Prop. 4.4 | BETA_FP_M §4 | none for BETA_FP_M; H_VERIFY §5 refuted old step | open / argued | 4 |
| V1 | R2 | U3_VERIFY C4 (order $p^2$) | open | 5 |
| (G2) | BETA_FP_LEMMA4.3 | — | formal | 6 |
| $\beta_j(g)$ independent of $K$ | LEMMAS_DE_VERIFY §2 item 2 | v1 verifier Point 7 | open | 9 |
| (H0), uniform in $j$ and history | CMP 109 (1.22) | BALABAN_CHECK item 10 | [Balaban], odd $L > 11$ | 7 |
| L5 history form and domain | LEMMA_DPRIME Rem. 3; LEMMAS_DE_VERIFY §2.1; v2_VERIFY §F | — | open | 8 |
| sweep reading | SWEEP §7 | LEMMA_DPRIME_VERIFY Steps 4–5 (rows 11, 12, 16, 18) | proved, unverified | 10 |
| odd $L$; $L > 11$ in Balaban | CMP 109 pp. 251–252 | v1 verifier Step 6 | input | 11 |
| SU(2) | — | — | scope | 12 |
| Lean D″ (one-argument) | KAPPA §5.3 | lean/README.md, `lemma_DDoublePrime` | Lean | §2.2 |
| Lean history form | — | — | open | 13 |
| Route B: Theorem B logic | BETA_FP_PW §3 | PW_VERIFY §6 items 1–4 | proved+verified, conditional (tag decision, PLAN.md) | B8 |

| | | | | |
|---|---|---|---|---|
| Route B: (U) | BETA_FP_PW §1.2 | PW_VERIFY §3 | hypothesis; $\Leftarrow$ (T1) + (H2) (BETA_FP_U); proved, unverified | B1, B2′ |
| Route B: (T1) | BETA_FP_U §2.3 | none | argued | B1 |
| Route B: $(B\theta) = \gamma_{rec} + \mathfrak{d}_F$ | BETA_FP_PW §3.3; BETA_FP_U §3 | PW_VERIFY §6 (Theorem B logic); none for BETA_FP_U | $\gamma_{rec}$ bounded given Thm 2.4 (proved, unverified); (DF) argued | B3 |
| Route B: A6 ($\beta$ identification, uniform in $k$) | SOURCES_D1_VERIFY 8; PW_VERIFY §3; BETA_FP_U §4.2 | | argued | B5 |
| Route B: $\theta_* < 1$ | PW_VERIFY §3 | — | trivial condition | B7 |
| LW (5.7) | cited | PW_VERIFY §3 (regime correct) | cited | B4 |
| BALABAN_CHECK item 11 readings | BALABAN_EXCERPTS | | unchecked | 10 |

## 11 Citation and independence

Related work 1. E. Sussman and L. Alpöge (with Claude; README: "written largely by LLM ... Novel mathematical ideas should be attributed to Claude (under the guidance of LA)"), "The Yang–Mills continuum limit along Bałaban's renormalization group: a complete account of Theorem 0" (LaTeX, files `theorem_0_complete_account_raw_{master,I,II,III}.tex`, 424k lines), and "SU(2) Yang–Mills theory in four dimensions via Bałaban's renormalisation group: construction, axioms and non-triviality" (`ym4_manuscript_zeroth_draft.pdf`, dated 6 October 2026) [23, 22]. Repository: `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, commit `1b92bd0347e3b8763cd8ecc1f96dc3c2a92dd310` (2026-10-27T05:45:28-05:00), accessed 2026-10-27. Repository created 2026-10-06. Disclaimer in README: "This repository contains work-in-progress, written largely by LLM. Its content has not been satisfactorily vetted. All claims should be understood as tentative, pending verification." Suggested keys: SA26-T0 (account), SA26-YM4 (draft).

Related work 2. OpenAI, "The canonical massive continuum limit of the two-dimensional O(3) model" (October 4, 2026), `github.com/openai/math`, family 215 (preprints folder; read locally from the PDF, `reports/ONLINE_SWEEP.md` §1.2; text in `reports/o3_continuum.txt`) [18]. Suggested key: OAI26-O3.

*How content from these works reached this project.* The record (file times, `PLAN.md`, read lists; `THEOREM2_PROOF_v2_VERIFY` §I) shows the following.

1. This chapter was written without reading the Sussman–Alpöge repository. Only the citation block above (`reports/SUSSMAN_ALPOGE_MAP.md` §5.1) was read.

2. First channel. On the night of 2026-10-27, `reports/ONLINE_SWEEP.md` §2 (about 20:20) read the Sussman–Alpöge repository directly. It reported that they prove a weak two-sided flow

with an additive constant, never need per-step positivity, use a first-passage cutoff, and invoke only containment $g_k \le \gamma$. This is the first project file with Sussman–Alpöge content.

3. Second channel. `reports/SUSSMAN_ALPOGE_MAP.md` (about 20:28) was written by a separate agent under a firewall. Through it, the bounded-$\kappa$ idea entered the project: the weak form of Theorem 2 needs only $\kappa_k$ bounded, not convergence (their Lemma 4.19 structure; `PLAN.md` line 146). `KAPPA_BOUNDED.md` (about 20:39) received this idea through its prompt. Its read list excludes the map and ONLINE_SWEEP.

4. Lemma D′ and the cross-scale sweep predate any Sussman–Alpöge content in project files. The shooting-without-positivity lemma and the band with additive constant appear in `THEOREM2_PROOF_VERIFY` Point 5(b) (about 20:05) and `LEMMA_DPRIME.md` (about 20:10). `LEMMA_DPRIME_SWEEP.md` (about 20:16–20:23) overlaps ONLINE_SWEEP in time, but its read list is LEMMA_DPRIME and its verifier only; it did not read ONLINE_SWEEP. Lemma D″ is in `KAPPA_BOUNDED.md`, written after the map.

5. Theorem K. Its result form, $|S_k - ck \log L| \le \kappa_*$, matches Sussman–Alpöge Lemma 4.2(b), $|\sum_l \beta_l^0 - kc_0| \le c_2$ (as reported in `PLAN.md` line 146; cited there, not proved). The proof in this project (Prop. 1.1, Lemma 2.1, Prop. 2.2 of `KAPPA_BOUNDED`) was written by firewalled agents from Balaban's papers and project files.

6. O(3) paper. Route B's doubling comparison follows the template of OAI26-O3 Lemma 6.1 (doubling difference; `BETA_FP_PW` uses the "O(3) Lemma 6.1 form"). The idea of a coupling band in place of per-step positivity also appears there (ONLINE_SWEEP §1.2(a)). `BETA_FP_PW` read ONLINE_SWEEP §1.2, §5, §8 and o3_continuum Lemma 6.1, and did not read ONLINE_SWEEP §2.

7. Route B (other than the O(3) template), the numerics, and the symbol results of §3.5 and §5 were derived by firewalled agents from Balaban's papers and project files.

Any write-up must cite SA26-T0, SA26-YM4, and OAI26-O3, and state these overlaps.

# A Change log (v2 → v3)

1. Range of $L$: new paragraph after the tag legend; §1.1 wording "Balaban assumes $L$ odd, $L > 11$, for the whole paper"; Theorem 2w stated for odd $L > 11$, or odd $L \ge 3$ with (H0) at this $L$ as a hypothesis; same restriction noted in §1.4 necessity, §2 notation, §4, §4.1 (H0), §6 order of constants and "what this does not give", §9 [Balaban], §10 items 7 and 11. "his constants need odd $L > 11$" replaced (`v2_VERIFY` §D, §L(7)).

2. L5 sharpened in §2.4, §2.1–§2.2 remarks, §6, §9, and §10 item 8: the clamped shooting evaluates $\beta$ on histories outside Balaban's domain; the nested-interval variant of Lemma D″ on true trajectories is described and tagged [not written] (`v2_VERIFY` §F).

3. (H″) ⇒ (B2) tagged [argued] in §5.3, §6, §9, §10 (`v2_VERIFY` §C).

4. "independent" / "disjoint" replaced by "formally distinct" in §4 title, §4 intro, §4.3; Route B points added: $\theta_* < 1$ (§4.2 statement, B5), $\beta$ identification uniform in $k$ (§4.2, §4.3 table row moved, B4), (U) plausibly needs the same uniform first-order vertex smoothness as $(C1\sigma)$ (§4.2, §4.3, B1) (`v2_VERIFY` §G).

5. §11 completed: ONLINE_SWEEP §2 named as the first channel; map named as the channel of the bounded-$\kappa$ idea; Lemma D′ and the sweep dated before any Sussman–Alpöge content; Theorem K's result form compared with their Lemma 4.2(b); OpenAI O(3) paper (family 215) cited for the doubling difference and the coupling band (`v2_VERIFY` §I).

6. §10: added $K$-independence of $\beta_j(g)$ (item 9), $\theta_* < 1$ (B5), B2_TWOSTEP as an unverified candidate proof of $(C1\sigma)$ (item 1; also §5.3, §9), unchecked Balaban items (item 10: CMP 119 p. 256 "other constants", $\beta_0$ before $\gamma$, `BALABAN_EXCERPTS_CHECK` item 11); hypothesis table of `v2_VERIFY` §K added; items renumbered (`v2_VERIFY` §J, §K).

7. §2.3: "4 of 19 table rows verified (rows 11, 12, 16, 18)" replaces "5 of 19 items"; sweep items 4, 6, 9, 14 added to the bullet list (`v2_VERIFY` §E).

8. §1.2: "D-free" replaced; $\gamma_*$ depends on $R$ through $B'$ (`v2_VERIFY` §A).

9. Status tags: `BETA_FP_PW_VERIFY.md` checked 2026-10-27; it has §1–§5 and no grade. Route B stays [proved, unverified]; "verifier running" replaced by "no grade". "Theorem 2w ⇒ Theorem 1" stays [proved, unverified], now also conditional on L5 (`v2_VERIFY` §N).

10. Smaller wording fixes (`v2_VERIFY` §A, §B, §E, §J, §L(9)): title no longer says "verified route"; §2.1 cites `LEMMA_DPRIME_VERIFY` grades 1, 2, 5 and notes the CORRECTED grades 3, 4, 6 of Step 7; (B2w) tag cites `KAPPA_BOUNDED_VERIFY` §9(iv); (h) and (H″) numerics tagged [numerical, run by the verifier]; B2_TWOSTEP no longer described as nonexistent; §9 notes the other Lean files.

(v3 status line: complete 2026-10-28, not verified as a whole document.)

# B Change log (v3 → v4)

1. §5 rewritten around (H2): definition (§5.3); Theorems C1 and B2 tagged [proved+verified, conditional on (H2)] (`B2_TWOSTEP_VERIFY` grades 1, 3, 4); threshold $\rho_2 < L$ with no strip loss; (H2) weaker than (H″); one-step sup-norm $C^1$ recursion fails at $L = 3$ (factor 1.005–1.033, §5.2); numerics table for $L = 3, 5, 7, 9$ (§5.4: 2.0634, 2.108, $\geq$ 2.1131, $\geq$ 2.1142 at $j = \infty$; margin 31% at $L = 3$, 58% at $L = 5$; $L$-trend rate claim corrected); one-step sufficiency at $L \geq 5$ (§5.5: $\rho_1(j = 1) = 1.860$, $\rho_1^2/L = 0.771$ at $L = 5$); certification plan and $j_0 \approx 8$ ($L = 3$), $\approx 6$ ($L = 5$), with the verifier's corrected inputs (§5.6); remaining inequality = (H2) (§5.7). The v3 text "B2_TWOSTEP: no verifier file, nothing used" is removed.

2. §1.3, §4 intro, §4.1, §4.3, §6: (B2) ⇐ (H2) stated; both routes rest on (H2) (`BETA_FP_U` §3.5, §4.3). New §4.2a: Route B inputs after `BETA_FP_U` ((U) ⇐ (T1) + (H2); $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$; (DF); Corollary 4.1; A6), tagged [proved, unverified; no verifier file].

3. §4.2, §4.3, §9, §10 B8: `BETA_FP_PW_VERIFY` now has §6 grades (items 1–4 CONFIRMED, 5–6 CORRECTED); v3's "no grade" is outdated.

4. §2.2 and §9: Lemma D″ is formalized in Lean (`YmLean.lemma_DDoublePrime`, no sorry), with (c′) constants $\beta = 3b/4$, $B' = 5b/4{+}2R$, $D' = 2R$ and (d) lower constant $b/2$ for $n{-}m \geq 8R/b$. §2.2 title gains [Lean]; v3's "[open as Lean work]" removed; §10 item 13 reduced to the history form.

5. §8.1 extended to $L = 9$ ($\beta_1 = 0.833$ at $Lp = 0.5$, constant $\approx +0.02$) and $L = 11$ (0.850 at $Lp = 0.5$, constant $\approx -0.04$; $N_g = 8$ adequacy unchecked) from `NUMERICS` §7. §8.2: per-level $\beta_2(0) \approx 0.32$ with the symmetric-Nyquist values 0.33334, 0.31679, 0.32283, and the (U) test row. §8.3: $\rho_1$ at $L = 5$, $\rho_2$ pointer to §5.4, $a_*(L = 5) = 0.0107$.

6. §9 ledger: [Lean] D″ added; [proved+verified] Theorems C1, B2, `h2_scan` structure, tail-bound derivation added; [proved, unverified] `BETA_FP_U` items and one-step sufficiency added, the `B2_TWOSTEP` candidate line removed; [argued] (T1), (DF), A6, $C^2/C^3$ under (H2) added; new block "corrected by verifiers since v3"; failed one-step method noted.

7. §10: new item H, (H2) as the single uniform hypothesis for both routes, status [numerical at $L = 3, 5, 9$; certification open]; item 1 reduced to H; Route B set rewritten per `BETA_FP_U` as (T1), (H2), (DF), LW (5.7), A6, (H0), $\theta_* < 1$; (DF) replaces $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ for this route; item 11 numerics range updated; item 13 reduced; hypothesis table updated.

8. Not changed: §1.1, §1.2, §1.4, §2.1, §2.3, §2.4, §3, §7, §11, and all other tags. `KAPPA_BOUNDED_VERIFY`'s (B2) obstruction statement is in its §7 and §9(vi) (the task brief cited §6; §6 there is the Lemma D″ check). `BALABAN_EXCERPTS_CHECK` has no change since v3.

9. *Tag change (deliberate; differs from the v4 Markdown).* The v4 Markdown keeps Route B (Theorem B of `BETA_FP_PW`) at [proved, unverified]. This version tags the logic of Theorem B as [proved+verified, conditional], citing `BETA_FP_PW_VERIFY` §6 items 1–4. Source: the line "TAG DECISION" in `PLAN.md` (2026-10-28). The inputs (U), (DF), (T1) from `BETA_FP_U` stay [proved, unverified]. Applied in §1.3, §4.2, §4.2a, §4.3, §6, §9, §10 (B8) and the hypothesis table; the Markdown sentences "the tag is kept as in v3" and "a tag change is left to the next review" are replaced accordingly.

Status: complete (2026-10-28). Not verified as a whole document.

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